

Blame Attribution and Disclosure Propensity

Jason D. Schloetzer
McDonough School of Business, Georgetown University
jds99@georgetown.edu

Ayung Tseng
Kelley School of Business, Indiana University
atseng@indiana.edu

Teri Lombardi Yohn*
Goizueta Business School, Emory University
tyohn@emory.edu

Yeo Sang Yoon
Carlson School of Management, University of Minnesota
yo0080@umn.edu

Running Head: Blame Attribution and Disclosure Propensity

* Corresponding author. We appreciate the comments of an anonymous reviewer, Mark Bagnoli, Ray Ball, Greg Clinch (editor), Ningzhong Li (discussant), Wei Li, Shiva Rajgopal, Lakshmanan Shivakumar, Jake Thomas, Jim Wahlen, and Frank Zhang. We thank the Institute for Crisis Management for providing business crisis classifications, senior members of the Crisis Communications group of Edelman for sharing their insights into how firms collaborate with crisis communication consultants to develop disclosure strategies in the wake of negative events, Colleen Honigsberg, Don Langevoort, and Neel Sukhatme for providing a legal view of negative event-related disclosure decisions, Craig Eich and Congci Hao for research assistance, the workshop participants at Indiana, Kent State, National University of Singapore, Purdue, Singapore Management University, and Yale, and participants at the 2017 Conference on the Convergence of Financial and Managerial Accounting Research and the 2017 HKUST Accounting Research Symposium. Jason Schloetzer appreciates the financial support of the William and Karen Sonneborn Term Associate Professorship.

Blame Attribution and Disclosure Propensity

ABSTRACT

We hypothesize that firms are less likely to disclose information regarding a material negative economic event for which the firm is likely to be blamed than for a negative economic event for which the firm is likely to be perceived as blameless. We identify 383 material negative economic events (casualty accidents, oil spills, catastrophes, investor class action lawsuits) and find that firms are approximately four times less likely to disclose information following a negative blamed event than for a negative blameless event. Consistent with the assertion underlying our hypothesis that disclosure of a blamed event increases the firm's reputation and litigation risks, we find that firms that disclose after a blamed event, but not a blameless event, experience greater reputation and litigation costs than firms that do not disclose. We also find that blame attribution provides incremental information over manager career concerns in the disclosure decision. These findings suggest that an event-specific factor—blame attribution—affects firms' propensity to provide disclosures about material negative economic events.

Keywords: economic event, voluntary disclosure, blame, litigation risk, firm reputation

JEL Classification: K22, M41

Data Availability: the list of 383 material negative economic events is available upon request

I. INTRODUCTION

While a vast accounting literature provides evidence that firm characteristics affect disclosure choices, less attention has been given to whether characteristics of the underlying economic event itself influence the firm's propensity to issue disclosures about the event. In this paper, we investigate whether firms are less likely to disclose information regarding negative economic events for which the firm is likely to be "blamed" than regarding negative economic events for which the firm is likely to be perceived as blameless.¹ Blame attribution theory suggests that individuals generally assign responsibility for a negative event to internal predispositions or external forces, but not both (Shaver 1985). Because of this behavior and because blamed events result in substantial reputation and litigation costs for the firm, we argue that firms consider whether the firm is likely to be blamed for a negative economic event when making disclosure decisions about the event.

We define a "blamed" event as a material negative economic event for which the firm is likely to be perceived as responsible for causing and hypothesize that firms are less likely to disclose information about blamed events than blameless events due to concerns about reputation and litigation risks. We identify material negative economic events that are observable through public sources and examine whether blame attribution for the event influences the firm's subsequent event-related disclosure decisions.²

Our sample of negative events includes catastrophes, major casualty accidents, and large oil spills identified by the National Weather Service, the National Transportation Safety Board,

¹ For example, BP and Chevron issued multiple detailed disclosures after Hurricanes Katrina and Rita that discussed the extent of oil spills caused by these catastrophes, yet both firms chose not to disclose information about oil spills of similar or even larger size that were caused by equipment failures or human error.

² Existing studies on asymmetric earnings news (Skinner 1994; Kothari, Shu, and Wysocki 2009) use the frequency of or the market reaction to positive news disclosures as a benchmark to estimate whether firms withhold negative news. In contrast, we identify the actual date of occurrence for only negative economic events from sources other than company disclosures and examine whether the negative event is subsequently disclosed by the affected firm.

and the Bureau of Safety and Environmental Enforcement accident reports, and the initial filing of investor class action lawsuits identified from the Stanford Securities Class Action Clearinghouse. The restriction to negative events allows us to extract from firms' asymmetric disclosure incentives between good and bad news (e.g., Skinner 1994; Kothari et al. 2009). Our focus on publicly observable events instead of firms' self-reported events mitigates the potential selection bias related to firms' disclosure behavior, such as firms' differential propensity to report positive versus negative events or bundling disclosure of favorable and unfavorable news (Bliss, Partnoy, and Furchtgott 2018).³ We further restrict our sample to *material* events, which are defined as events that significantly affect investors (Heitzman, Wasley, and Zimmerman 2010), employees, customers, or suppliers (Donaldson and Preston 1995).⁴ Our final sample includes 191 sudden accidents (61 catastrophes, 41 casualty accidents, and 89 oil spills) and 192 investor class action lawsuits from 2001 to 2013.

We collect company disclosures from SEC filings and press releases based on the conjecture that disclosures for investors are likely to be disseminated via SEC filings whereas disclosures for employees, customers, and suppliers (as well as investors) are likely to be disseminated via press releases. We read through all SEC filings and press releases for the 30 days following the first occurrence of an event to identify the firm's event-specific disclosures.⁵ For the sudden accident sample, our empirical proxy for blame attribution is whether the accident is caused by an internal or external force. We assume that internal forces, such as human error, are more

³ Ben-Rephael, Da, Easton, and Israelsen (2019) document a greater market reaction to economic events than to the 8-K filings, suggesting that market participants are aware of economic events prior to firm disclosure about the event.

⁴ We identify material events based on the stock price reaction for catastrophes, the occurrence of death or substantial damage for casualty accidents, the number of barrels spilled for oil spills, and lawsuits filed in federal court for class action lawsuits alleging overly optimistic forward-looking statements.

⁵ We note that this direct identification of event-specific disclosures is distinct from the approach of counting the total number of SEC filings or press releases as a proxy for disclosure quality (e.g., Balakrishnan, Core, and Verdi 2014; Burks, Cuny, Gerakos, and Granja 2018; Bird, Karolyi, and Ruchti 2017).

likely to be perceived as blamed compared to external forces, such as weather, because firms are perceived to have more control over internal forces. For the investor class action lawsuit sample, our empirical proxy for blame attribution is whether or not a lawsuit is subsequently dismissed by the court. Since our lawsuit sample is restricted to a single claim of overly optimistic forward-looking statements, we assume that firms know whether these optimistic statements were *intended* to manipulate public perception and therefore have insight into whether the lawsuit is likely to be dismissed. Therefore, blame attribution is captured by the subsequent court dismissal decision.

We find that, among the 383 material negative economic events, 35 percent are associated with firm disclosure within 30 days after the event and 50 percent are associated with firm disclosure within 90 days after the event. Our multivariate tests reveal that firms are *at least four times less likely* to release information following a material negative event for which the firm is likely to be blamed relative to a similar negative event for which the firm is likely to be perceived as blameless. Throughout our analyses, we control for firm characteristics documented in prior research to be associated with firm disclosure choices.⁶ We also control for the materiality of the event by including four measures of materiality in the analyses: the cumulative stock return in the days around the initial occurrence of the event, the change in the implied volatility of stock returns around the event, the number of news articles about the event, and the average sentiment of the event-specific news articles. We show that including the four event materiality measures improves the explanatory power of the model by 18 percent over the model that includes firm characteristics. More importantly, including blame attribution improves the explanatory power of the model by 25 percent over the model that includes firm characteristics and the four event materiality measures.

⁶ We include 25 disclosure determinants to identify the incremental impact of blame attribution on the disclosure propensity, incorporate year and industry fixed effects, and cluster standard errors by firm to capture unobservable factors.

The statistical significance and economic magnitude of blame attribution results are robust across the different types of negative events and to additional analyses. Further, prior research suggests that, due to agency problems between the manager and the shareholders, managers' disclosure decisions are associated with their career concerns (Kothari et al. 2009). We find that blame attribution is incremental to managers' career concerns in explaining the firms' propensity to disclose after a material negative event.

Our hypothesis is based on the assertion that disclosure of a blamed event increases the firm's reputation and litigation risks associated with the event. To provide evidence on the validity of this assertion, we also examine whether disclosure of a blamed event is associated with higher reputation and litigation costs. To do so, we use the Fortune 1000 Most Admired Companies score and the RavenPack Event Sentiment Score to proxy for firm reputation, and we use the lawsuit duration and the lawsuit settlement amount to proxy for litigation costs. We find that following a blamed event, disclosing firms experience larger declines in reputation and higher litigation costs than non-disclosing firms. However, for firms experiencing a blameless event, disclosure does not increase reputation and litigation costs. These findings are consistent with our assertion that firms are less likely to disclose information about a blamed event than a blameless event due to concerns about the effect of disclosure on the firm's reputation and litigation risks.

This study contributes to the accounting literature in several ways. Our primary finding of a relation between blame attribution for a material negative economic event and the propensity for a firm to disclose about the event contributes to research on event-specific factors and firm disclosure. For example, Amir, Levi, and Livne (2018) use the equity market reaction and disclosure timeliness to distinguish between cyber-attacks that firms voluntarily disclose and cyber-attacks that firms are forced to disclose. Barth, McNichols, and Wilson (1997) examine how

environmental regulations and estimation uncertainties affect firm disclosure about environmental liabilities. Research also examines the consequences of event-specific disclosures (e.g., Christensen, Floyd, Liu, and Maffett 2017; Cho, Frankel, and Martin 2019; and Lee, Hutton, and Shu 2015). Our study contributes to this literature by examining a broad spectrum of negative economic events and determining whether an event characteristic—blame attribution—impacts the firm’s choice to disclose.

Our study contributes to research on attribution theory and supplemental statements in disclosures. Prior research (Baginski, Hassell, and Hillison 2000; Baginski, Hassell, and Kimbrough 2004) suggests that supplementary explanations in firm disclosures are consistent with attribution theory (Weiner 1985, 1986) such that firms use disclosures to manipulate public perception of the cause of negative news.⁷ We contribute to this literature by applying attribution theory (Shaver 1985) to the more fundamental decision of whether a firm discloses about a negative event, rather than to the information that is conveyed within a disclosure.⁸

Our study also contributes to research on disclosure timeliness and the sign of news. Prior research suggests that firm litigation and reputation risks (Skinner 1994; Kothari et al. 2009; Marinovic and Varas 2016) and managers’ career concerns (Kothari et al. 2009) affect the timeliness of firm disclosure of good versus bad news. We contribute to this literature by identifying material negative economic events and examining the firm’s decision to disclose the event rather than the timeliness of good versus bad news disclosures. That is, we examine only negative events, rather than comparing bad news to good news, and examine whether a

⁷ Further, Barton and Mercer (2005) and Kimbrough and Wang (2014) examine how the plausibility of such managerial-attributing behavior influences financial analysts’ forecasts and equity market reactions.

⁸ In subsection “Other Disclosure Characteristics,” we also examine the characteristics of the disclosures provided.

characteristic of the negative event—blame attribution—is associated with the decision to disclose, rather than the timing of the disclosure.

Our blame attribution hypothesis and findings also contribute to understanding how reputation and litigation risks affect firms' disclosure decisions. While prior research suggests that timely disclosure of bad news *decreases* the firm's reputation and litigation costs associated with *disclosure* (Skinner 1994; Kothari et al. 2009; Marinovic and Varas 2016), our findings suggest that disclosure of blamed events *increases* the firm's reputation and litigation costs associated with the negative *event* itself.

The remainder of the paper proceeds as follows. In Section II, we develop our hypothesis. In Section III, we describe our sample and variable definitions. Sections IV and V provide the results of our empirical tests, and Section VI concludes the paper.

II. HYPOTHESIS DEVELOPMENT AND RELATED RESEARCH

Blame Attribution Hypothesis

GAAP, SEC rules, and securities laws require firms to disclose material events, although they provide firms with a degree of discretion over which events require disclosure.⁹ In this study, we examine firms' decisions regarding whether or not to disclose about a material negative event. Disclosure theory suggests that firms consider proprietary costs associated with disclosure

⁹ For instance, FASB's definition of materiality is: "Information is material if omitting it or misstating it could influence decisions that users make on the basis of the financial information of a specific reporting entity. In other words, materiality is an entity-specific aspect of relevance based on the nature or magnitude or both of the items to which the information relates in the context of an individual entity's financial report. Consequently, the Board cannot specify a uniform quantitative threshold for materiality or predetermine what could be material in a particular situation..." FASB SFAS No. 8 (2010). The U.S. Supreme Court also provides a broad-based definition: "[materiality] is inherently fact-specific, depending upon whether a 'reasonable investor' would have viewed the relevant information as having significantly altered the total mix of information made available." *Matrixx Initiatives, Inc. v. Siracusano*, No. 09-1156, 2011 WL 977060 (U.S. Mar. 22, 2011). SEC rules reinforce the notion that firms have a degree of discretion over which events to disclose via Form 8-K, as 8-K Item 8.01 states: "The registrant may, at its option, disclose under this Item 8.01 any events, with respect to which information is not otherwise called for by this form that the registrant deems of importance to security holders."...SEC (2004). These definitions stipulate that firms are required to disclose material economic events, but the ambiguity in the definitions creates debate regarding which events require disclosure.

(Verrecchia 1983). We argue that firms' proprietary costs of disclosure about material negative events differ depending on blame attribution for the event.

Blame attribution theory suggests that individuals generally assign responsibility for a negative event to internal predispositions or external forces, but not both (Shaver 1985). Because of this behavior and because events for which the firm is likely to be responsible for causing result in substantial reputation and litigation costs for the firm, we hypothesize that firms are less likely to disclose information regarding a blamed event than a blameless event due to concerns about the effect of disclosure on the firm's reputation and litigation risks related to the event.

Reputation Risk with Respect to Blame Attribution

Firms may be less willing to disclose information about a blamed event relative to a blameless event because of the impact of disclosure on the firm's reputation risk. Prior research defines reputation as the perception of the firm held by its stakeholders (e.g., investors, employees, customers, and suppliers), and reputation risk as a potential decrease in reputation which may affect future actions of stakeholders toward a firm (Kreps and Wilson 1982; Fombrun and Shanley 1990; Dutton, Dukerich, and Harquail 1994; Brown and Dacin 1997; Walker 2010). This risk arises because stakeholders use reputation as an imprecise signal to assess firm attributes that are difficult to observe (e.g., process quality, corporate culture; Fombrun and Riel 1997). We argue that the release of information regarding a blamed event decreases the firm's reputation more than the release of information about a blameless event. In other words, stakeholders are more likely to downgrade a firm's reputation after the disclosure of detailed facts regarding a blamed event than after disclosure of facts regarding a blameless event. Consistent with this perspective, Perry and Fontnouvelle (2005) find that reputational loss, measured as the market reaction to an operational

loss announcement, is greater following an operational loss caused by internal forces (i.e., a blamed event) than by an operational loss caused by external forces (i.e., a blameless event).

Litigation Risk with Respect to Blame Attribution

Firms are likely to consider the threat of litigation (e.g., local communities filing a lawsuit against a firm after an oil spill) after a negative event as well as the effect of disclosure on the likelihood and outcome of any related litigation. A stream of legal literature notes that it is often challenging for plaintiffs to demonstrate that the defendant knew, or should have known, that her actions were wrong (i.e., scienter) at the start of the lawsuit before having access to witnesses and internal documents during a trial (Pritchard and Sale 2005; Honigsberg, Rajgopal, and Srinivasan 2019). Facts released via defendants' pre-trial public disclosures potentially help plaintiffs plead that the defendant acted with scienter. We argue that whether firms are willing to disclose likely relates to their perception of blame attribution for the event because the litigation risk related to disclosed facts is potentially greater for a blamed event than for a blameless event. It then follows that firms might be hesitant to issue disclosures following a blamed event in order to minimize the public information that can be used against them in litigation.

Even after a lawsuit has been filed, blame attribution and disclosure choices are likely to affect the remaining litigation risk with respect to the duration of the suit and the amount of the expected damage. Legal scholars recognize that significant information asymmetry remains between a plaintiff and a defendant after a lawsuit is initially filed, and studies examine how this information asymmetry influences firms' disclosure decisions. For example, firms may attempt to settle with plaintiffs to avoid trial because firms are concerned about the need to disclose information during the discovery phase (Grundfest and Huang 2006). In addition, the Private Securities Litigation Reform Act of 1995 prevents plaintiffs from accessing witnesses and internal

documents before the resolution of motions to dismiss. Thus, defendants' public disclosures (e.g., press releases and SEC filings) become crucial sources of information that plaintiffs can use to prove key elements of their claims (Pritchard and Sale 2005). This implies that defendants might be cautious about the disclosure of facts that could be used by plaintiffs.

Blame Attribution Hypothesis and Related Literature

We highlight how our blame attribution hypothesis for firms' disclosure decisions after negative economic events relates to hypotheses offered in prior work on firm disclosure behavior. Verrecchia's (1983) proprietary cost hypothesis argues that a firm's disclosure decisions are influenced by the potential use of disclosed information by stakeholders. Our blame attribution hypothesis extends the proprietary cost hypothesis by suggesting that firms' proprietary costs of disclosure are higher for blamed negative economic events than for blameless events. That is, for blamed events, firms consider that its disclosures are more likely to be used by plaintiffs to prove their claims and by stakeholders to lower their perceptions of the firm's reputation. We therefore argue that firms' proprietary costs of disclosure regarding a negative event differ depending on the blame attribution for the event.

Our blame attribution hypothesis also extends the litigation and reputation hypotheses in the prior literature on the timing of disclosure of good versus bad earnings-related news. Specifically, this stream of research argues that firms face an increase in reputation and litigation risk related to disclosure if they *do not* disclose negative earnings news in a timely manner (Skinner 1994; Kothari et al. 2009; Marinovic and Varas 2016). By contrast, our blame attribution hypothesis suggests that firms face an increase in reputation and litigation risk related to the event if they *do* disclose information about a material negative economic event for which the firm is likely to be blamed. That is, while the prior literature suggests that firm disclosure *decreases* the

reputation and litigation costs associated with lack of *timely disclosure*, the blame attribution hypothesis suggests that firm disclosure *increases* the reputation and litigation costs associated with the *event* itself for blamed events. Our blame attribution hypothesis therefore considers the effect of disclosure on the reputation and litigation costs related to the event instead of those related to lack of disclosure.

Lastly, our blame attribution hypothesis extends the manager career concern hypothesis related to disclosure (Kothari et al. 2009). The career concern hypothesis assumes self-interested managers and is based on the agency problem between the manager and shareholders. It posits that managers delay bad news relative to good news because disclosure of bad news might lead to poor career outcomes (e.g., termination, lower compensation). Therefore, the career concern hypothesis assumes that the interests of the managers and the firm are misaligned. Our blame attribution hypothesis also assumes self-interested managers but does not assume that managers' and the firm's interests are misaligned. Rather, our hypothesis considers managers who act in the best interest of the firm and considers that the firm's proprietary costs of disclosure differ across blamed and blameless negative events. Therefore, the blame attribution and manager career concern hypotheses are distinct and both are likely to explain disclosure decisions depending on the situation.¹⁰

III. SAMPLE AND VARIABLE DEFINITIONS

Material Negative Economic Events

Our goal is to examine whether blame attribution for material negative economic events influences the firm's choice to issue event-specific disclosures. This examination requires a sample of negative events that are exogenous to, or unexpected by, the affected firm. It also requires

¹⁰ In subsection "Incorporating CEO Career Concerns," we present evidence that blame attribution provides incremental information over manager's career concerns in explaining the decision to disclose a negative event.

credible information sources other than company disclosures to identify the event and its characteristics. With these criteria in mind, and following guidance from crisis communication consultants, we identify four types of negative events that satisfy these criteria—catastrophes, casualty accidents, oil spills, and investor class action lawsuits.¹¹ We include these four types of events because we expect blame attribution for the event to affect disclosure across a variety of negative events. We note, however, that catastrophes, casualty accidents, and oil spills are more likely to be exogenous than class action lawsuits. We therefore examine the relation between blame attribution for the event and firm disclosure about the event for the pooled types of events as well as for subsamples of each type of event.

We further restrict our sample to material events, defined as events that significantly affect stakeholders' decisions. We identify material events based on the stock price reaction for catastrophes, the occurrence of death or substantial damage for casualty accidents, the number of barrels spilled for oil spills, and lawsuits filed in federal court for class action lawsuits alleging overly optimistic forward-looking statements. Stakeholders (investors, employees, customers, and suppliers) likely demand information following both material sudden accidents (i.e., catastrophes, casualty accidents, and oil spills) and investor class action lawsuits. Therefore, firms likely consider whether to issue disclosures following each of these events. In practice, firms frequently engage public relations consultants and legal counselors to help shape disclosure choices after a material negative event.¹²

¹¹ We appreciate the guidance provided by the Institute for Crisis Management (ICM), a prominent crisis consulting firm that has published an annual business crisis report since 1990.

¹² Premier public relations firms have consultants devoted to business crisis communications (e.g., <https://www.edelman.com/expertise/crisis-reputation-risk>). Similarly, major law firms often have practice areas that assist in shaping disclosures following a material negative event (e.g., <http://www.stepto.com/practices-200.html>).

Identification of Catastrophes

As catastrophes are rare events, we focus on years 2005 (due to Hurricane Katrina), 2011, and 2012 as our sample period because these years have more severe catastrophes. The identification of specific firms affected by a catastrophe is challenging because it typically affects many firms and their subsidiaries in a broad geographic region. For example, McKnight and Linnenluecke (2017) identify a sample of natural disasters and then rely on press releases to identify the affected firms. However, they acknowledge that this approach potentially creates a sample that is biased toward large firms and firms that are willing to provide disclosures.

Since our study aims to examine whether firms issue disclosures following the event, we do not use corporate disclosures to identify firms affected by the catastrophe. We instead identify affected firms based on the equity market reaction to the occurrence of the catastrophe. Specifically, we identify firms (1) whose stock price fell in the five trading days following a catastrophe two or more standard deviations below the mean firm-specific weekly return (following Kim, Li, and Zhang 2011) and (2) who did not issue any confounding scheduled disclosures (e.g., earnings announcement, management guidance, dividend announcement) during the five day period.¹³ Our final sample includes 61 publicly-traded firms materially affected by five catastrophes: 28 by the 2005 Hurricane Katrina, 19 by the 2011 Japanese Tsunami, 12 by the 2012 Hurricane Sandy, 1 by the 2011 Joplin Tornado, and 1 by the 2011 New Zealand Earthquake.

¹³ Since catastrophes captured by this approach are potentially more material than other sudden accidents identified by the number of deaths or the barrels of oil spilled, we include the first and second moments of the market reaction as control variables to account for heterogeneous degrees of materiality within the sudden accident sample. Moreover, we incorporate spill size, water depth, and distance to shore as additional proxies for materiality in a sample that includes only oil spills as a robustness test to address the heterogeneity issue.

Identification of Casualty Accidents

We develop our sample of casualty accidents from the NTSB's casualty accident reports and the FAA's aviation accident database.¹⁴ When comparing the NTSB accident reports to the FAA aviation accident database, we find that NTSB writes reports on more severe casualty accidents in terms of the severity of injuries, the number of deaths, and the degree of damages to facilities. Therefore, we restrict our search of the FAA aviation database to the following: 1) accidents involving deaths and substantial damages to aircrafts, or 2) accidents leading to destroyed aircrafts. Our final sample includes 41 publicly traded firms affected by casualty accidents during 2005, 2011, and 2012. Among these 41 material negative events, 18 are aviation accidents, 15 are railroad accidents, 6 are marine accidents, and 2 are pipeline accidents. Two marine accidents occurred internationally, and three casualty accidents involved more than one firm. Twelve accidents occurred in 2005, 13 in 2011, and the remainder in 2012.

Identification of Major Oil Spills

We use the BSEE's oil spill summary reports to identify severe oil spills, defined as those spills involving greater or equal to 50 barrels, across three domestic regions: the Gulf of Mexico, Alaska, and the Pacific.¹⁵ Our final sample includes 89 material oil spills by publicly-traded firms during the period from 2001 to 2013.

Identification of Class Action Lawsuits

We use the Stanford Law School's Securities Class Action Clearinghouse (SCAC) to identify class action lawsuits filed in Federal Courts. Of note, we are careful to restrict the investor

¹⁴ These reports are available at <http://www.nts.gov/investigations/AccidentReports/Pages/AccidentReports.aspx> and <http://www.nts.gov/layouts/nts.aviation/index.aspx>.

¹⁵ Reports are available at <https://www.bsee.gov/site-page/spills>. Different from our approach, Barth et al. (1997) examine disclosure choices within publicly-traded firms named as potentially responsible parties to clean up Superfund sites.

class action sample to lawsuits that *only* claim overly optimistic forward-looking statements. This approach allows us to hold constant the nature of litigation (i.e., a subset of 10b-5 lawsuits) and remove cases that are confounded with other allegations (e.g., accounting fraud and restatements, related party transactions, government investigation and violations) because different allegations potentially face different pleading standards (see evidence in Pritchard and Sale 2005).¹⁶ Our final sample includes 192 cases during 2005, 2010, 2011, and 2012.

Event-Specific Disclosure

For all firms included in our sample of material negative events, we read through all SEC filings and press releases for the 30 days following the event to identify event-specific disclosure.¹⁷ We rely on SEC filings and press releases because existing studies argue that firms use press releases to reach a broader group of stakeholders (e.g., Magness (2006) in the context of environmental accidents, Madsen and Rodgers (2015) and McKnight and Linnenluecke (2017) in the context of natural disasters). Untabulated descriptive statistics show that 76 percent of disclosed sudden accidents are released via press releases, while 93 percent of disclosed securities lawsuits are filed via SEC filings. Untabulated evidence also shows that more than half of firms that disclose release their initial disclosures within five days after the negative event and 80 percent of initial disclosures are released within two weeks following the event. This suggests that most firms that disclose event-specific information do so within 30 days. A robustness test discussed in

¹⁶ Investor class action lawsuit samples in prior studies often include all 10b-5 cases without distinguishing the nature of claims (e.g., Skinner 1997, among others). We read each 10b-5 case to identify a sample that is based only on claims of overly optimistic forward-looking statements. Field, Lowry, and Shu (2005) take a similar approach by removing cases that involve restatements. Moreover, Pritchard and Sale (2005) show that investor class action lawsuits alleging misleading forward-looking statements are less likely to be dismissed than other allegations. Our litigation sample is representative of the overall population because law firms that are present most frequently in our sample (e.g., Robbins Geller Rudman & Dowd, Brower Piven) are similar to those in a sample of all 10b-5 cases from the Stanford Securities Class Action Clearinghouse and overlap with the list of top securities class action law firms ranked by ISS based on settlement amounts.

¹⁷ Appendix A illustrates the importance of event-specific disclosures to stakeholders. Appendix B provides two examples of event-specific disclosures.

subsection “Disclosure within 90 Days” documents consistent evidence when we expand the post-event disclosure window to 90 days with a focus on 10-K/Q filings.

We create an indicator variable, *Disclose (30 days)*, that equals one if we identify any SEC filings or press releases mentioning the event in our sample within the 30 days following the event, and zero otherwise. For investor class action lawsuits, we further search for disclosures issued during the 90 days prior to the identified event date to capture instances in which firms might preempt the lawsuit filing date. We choose the 90-day window because firms would have issued at least one financial report, either a 10-K or a 10-Q, during this pre-event period. We find only one instance in which a firm discussed a potential lawsuit prior to its filing.

Proxies for Blame Attribution

We identify separate proxies to capture the firm’s perceived blame attribution for the sudden accident sample and the investor class action lawsuit sample.

Externally-Caused versus Internally-Caused Accidents

Our first proxy is based on the controllability of an event. Motivated by the management and public relations literatures (e.g., Coombs 2007), stakeholders are more likely to perceive a firm to be responsible for an event when it is caused by an internal force than an external force. This is because firms are perceived to have more control over internal forces than external ones. For example, a firm is more likely to be blamed for an oil spill caused by human error (an internally-caused event) than for an oil spill caused by a hurricane (an externally-caused event).

We examine a sample of 191 sudden accidents, which capture natural disasters, casualty accidents, and oil spills. We partition these events into internally-caused and externally-caused events.¹⁸ All 61 natural disasters are classified as externally-caused events. For 41 casualty

¹⁸ We exclude three sudden accidents that were attributable to both internal and external forces. After the exclusion, our final sample has 191 sudden accidents.

accidents, we read NTSB and FAA accident reports to assess whether each accident was caused by an internal or external force. For 89 oil spills, the 12 events caused by Hurricanes Katrina or Rita, and one event due to a lightning strike are classified as externally-caused events. Internal causes for casualty accidents and oil spills often relate to human error and equipment failures.

We create a dummy variable, *External*, that equals one for 104 externally-caused events and zero for 87 internally-caused events, as our proxy for whether the firm is likely to be perceived as relatively blameless (externally-caused) or blamed (internally-caused). The blame attribution hypothesis predicts that firms facing an externally-caused event are *more* willing to provide disclosure than firms facing an internally-caused event because external causes are unlikely to be under the firm's control and, hence, the firm is likely to be perceived as blameless.

Dismissed versus Non-Dismissed Lawsuit Cases

Our second proxy is based on whether an investor class action lawsuit is dismissed by the court. We use the lawsuit outcome, dismissed versus non-dismissed, to distinguish between litigation cases for which the firm is likely to be perceived as blameless or blamed, respectively. We include only those cases claiming that the firm made overly optimistic forward-looking statements to hold constant the nature of litigation. We create a dummy variable, *Dismiss*, which equals one for 108 dismissed cases (blameless) and zero for 84 cases that are not dismissed (blamed). All 84 non-dismissed cases are eventually settled between the defendant and the plaintiff in our sample.

Forward-looking statements are protected by the statutory safe harbor, so a plaintiff needs to prove that the defendant is “with actual knowledge of their falsity” (Pritchard and Sale 2005, p. 133).¹⁹ The statute requires that the plaintiff has to “state with particularity facts giving rise to a

¹⁹ Exchange Act §21E(c)(1)(B), 15 U.S.C. §78u-5(c)(1)(B).

strong inference that the defendant acted with the required state of mind.”²⁰ Our use of the court dismissal decision as a proxy for the firm’s perceived blame attribution for the event relies on two assumptions. First, the firm knows its own state of mind regarding whether the firm *intended* to mislead investors by providing overly optimistic forward-looking statements. Second, the estimated state of mind by the judge is positively correlated with the firm’s true state of mind. If judicial assessments contain substantial error, this empirical proxy may not reflect the firm’s perceived blame attribution and subsequently may not relate to the disclosure propensity.²¹

Empirical Model

We examine firms’ disclosure choices in the wake of a material negative event using the following empirical model:

$$\begin{aligned} Disclose(30\ days)_{i,e} &= \beta Blameless_{i,e} \\ &+ \sum \gamma_j Materiality_{j,i,e} + \sum \delta_l OtherDeterminants_{l,i,t-1} + \sum \alpha_k Year_k \\ &+ \sum \mu_m Industry_m + \varepsilon_{i,e} \end{aligned}$$

The dependent variable is *Disclose (30 days)*, an indicator variable for whether firm *i* issues any disclosure regarding the event within the 30 days following the event. We estimate logistic regressions because the dependent variable is an indicator. Our variable of interest is *Blameless*, an indicator variable that captures whether the firm is likely to be perceived as *blameless* regarding the negative event. Evidence consistent with our hypothesis would yield $\beta > 0$.

²⁰ Exchange Act §21D(b)(2), 15 U.S.C. §78t-5(b)(2).

²¹ Measurement error in the independent variable of interest, *Blameless*, could attenuate the coefficient estimate toward zero (Griliches and Ringstad 1970), weakening our ability to find evidence consistent with the blame attribution hypothesis.

We include four measures of event materiality. Since rules and regulations typically define materiality from an investor's perspective, we use the first and second moments of the investor reaction on the date that the event initially occurred to capture event materiality. The first moment is the cumulative market-adjusted return during the three days $[0, +2]$ around the event date, which captures the shift in investor expectations of firm value (*Three-day CAR*).²² The second moment is the change in implied volatility derived from option prices during the $[-3, +3]$ period around the event date, which measures the increase in investor uncertainty around the event (Δ *Implied volatility*). In addition to the investor reaction, we also include two event-specific media attention measures. We identify press releases or full articles in RavenPack regarding the event on the date the event initially occurred. *Media coverage* counts the number of event-specific articles and *Media sentiment* is the average sentiment score of the event-specific articles (Bushman, Williams, and Wittenberg-Moerman 2017).

To validate our materiality measures, we examine the correlations between our measures and the filing of an event-related 8-K as well as the lawsuit settlement amount. We find, untabulated, a marginally significant (two-tailed p-value of 0.14) positive correlation of 0.08 between *Three-day CAR* and the filing of an event-related 8-K. We also find a marginally significant (two-tailed p-value of 0.13) positive correlation of 0.17 between Δ *Implied volatility* and the settlement amount, and a significant (two-tailed p-value less than 0.01) positive correlation of 0.56 between *Media coverage* and the settlement amount. We view these positive correlations as consistent with the event-date investor reaction and media attention at least partially capturing the materiality of the event.

²² In subsection "Disclosure within 30 Days," we discuss alternate market-based materiality measures, such as using the absolute value or an extended window to calculate the market reaction.

We also control for determinants that have been shown in prior studies to relate to firms' disclosure choices (e.g., Leftwich, Watts, and Zimmerman 1981; Botosan and Harris 2000; Rogers and Stocken 2005; Verrecchia and Weber 2006; Brown and Hillegeist 2007; Khan and Watts 2009; Rogers, Skinner, and Van Buskirk 2009; Fu, Kraft, and Zhang 2012; Kim and Skinner 2012; Huang, Jennings, and Yu 2017; Huang, Hui, and Li 2019). Specifically, we include market capitalization to capture firm size, the market-to-book ratio to capture growth, return-on-assets and an indicator variable for a reported loss to capture firm profitability, leverage, and depreciation expense deflated by lagged total assets to capture the firm's investment cycle. We include past return momentum, past return volatility, firm age, an indicator for the firm being listed on the NYSE, and an indicator for the firm being incorporated in the U.S. We also include an indicator for the firm operating in a higher litigation risk industry (following Francis, Philbrick, and Schipper 1994), the estimated probability of the firm being sued based on the model in Rogers and Stocken (2005), the estimated probability of the firm being sued based on the model in Kim and Skinner (2012), and the estimated judge ideology from Huang et al. (2019) to capture firm's litigation risk.²³ We include industry competition based on the Herfindahl index, the frequency of confidential treatment requests, and the import tariff rate to capture the firm's proprietary costs of disclosure regarding competitors. We include an indicator for the firm issuing management guidance, the ownership percentage of insiders, the ownership percentage of institutional investors, the number of analysts following the firm, and the number of public debt offerings to capture the firm's information environment and quality of corporate governance. We include the bid-ask spread and analyst forecast dispersion as proxies for the information asymmetry associated

²³ We include all four of these proxies for litigation risk in our model because the correlations between them suggest that multicollinearity among the variables is not an issue. The highest correlation among the four litigation proxies is 0.23 between the Rogers and Stocken (2005) measure and the Kim and Skinner (2012) measure.

with the firm.²⁴ All variables are measured in the year prior to the event. Finally, we include year fixed effects and industry fixed effects based on the first digit of the SIC code. We cluster standard errors by firm.

IV. RESULTS

Summary Statistics

Table 1 reports that, for our sample of 383 negative events, 35 percent are associated with a disclosure within the 30 days following the event. The average *Three-day CAR* around the event date is -2 percent and the average increase in implied volatility is 1 percent. On average, there are two media articles covering the event on the event date and the average sentiment score is -0.40 (where the normalized sentiment score ranges from -1 to +1 with zero reflecting neutral news). In untabulated analyses, we find that the 383 negative events are distributed across all one-digit SIC industries, and are not clustered within any particular industry or concentrated within regulated industries.

Table 1 also reveals that *Disclosure CAR*, the three-day cumulative abnormal return (CAR) around each event-specific firm disclosure in the 30 days after the material negative economic event, is -1 percent, on average.²⁵ In untabulated analyses, we find that the sum of the three-day cumulative abnormal return (CAR) across all the firm's event-specific disclosures issued in the 30 days after a material negative economic event is -2 percent, on average.²⁶ The average sentiment score for the firm's disclosure for sudden accidents (lawsuits) is -0.52 (-0.58). In comparison, during the period from 2001-2013 for our sample firms, only 12 percent of disclosures or events

²⁴ Please refer to Appendix C for definitions of all control variables.

²⁵ When we limit the analysis to the 134 firms that disclose within the 30 days after the event, the average three-day CAR around the negative event is -0.3 percent, which is insignificantly different from -1 percent, the average three-day CAR around the firm's event-specific disclosures issued within the 30 days after the event.

²⁶ In untabulated analysis, we find evidence that press release disclosures issued within five days after a material negative event generate a more negative market reaction than those issued later in the 30-day period. However, we do not find a similar trend for disclosures disseminated through SEC filings.

have a sentiment score lower than -0.40. These statistics suggest that our sample events and the corresponding firm event-specific disclosures are important to the firms' stakeholders.

In untabulated analyses, we find that the 383 negative events are associated with 302 unique firms. Among the 302 firms, 269 firms experience only one event. Among the 33 firms with more than one event, 20 firms experience both at least one blamed event and at least one blameless event. Among these 20 firms, 12 firms follow a disclosure practice that is consistent with our blame attribution hypothesis (i.e., disclose blameless events and not disclose blamed events), five firms always choose no disclosure, one firm always discloses, and two firms adopt a practice that is inconsistent with our hypothesis.

Table 2 presents correlations among the key variables used in the analyses. Consistent with the blame attribution hypothesis, the correlation between *Disclose (30 days)* and *Blameless* is significantly positive, indicating a higher (lower) likelihood of disclosure for blameless (blamed) events. Moreover, the four event materiality measures are not significantly correlated with *Blameless*. The only exception is a positive Spearman correlation between *Blameless* and Δ *Implied volatility*, suggesting a greater increase in investor uncertainty around blameless events than blamed events. According to the uncertainty hypothesis in Dye (1985) and Jung and Kwon (1988), this positive association between uncertainty and blameless events suggests that firms experiencing a blameless event should be less likely to disclose. This prediction is counter to our blame attribution hypothesis, which suggests that firms with blameless events should be more likely to disclose.

Disclosure within 30 Days

In Table 3, we report the results of logistic regressions which examine whether the blame attribution associated with the economic event incrementally explains the firm's disclosure choice

beyond event materiality and firm characteristics. Model (1) is a benchmark model which includes 25 firm characteristics, includes industry and year fixed effects, and clusters standard errors by firm.²⁷ Model (2) adds the four event materiality measures, and Model (3) tests our hypothesis by adding the blame attribution proxy. The pseudo R-squared increases by 18 percent from 0.17 in Model (1) to 0.20 in Model (2). This improvement is attributable to the significant and positive coefficient on *Three-day CAR* (p-value less than 0.01) in Model (2).²⁸ More importantly, the pseudo R-squared further increases by 25 percent from 0.20 in Model (2) to 0.25 in Model (3). This improvement is attributable to the significant and positive coefficient on *Blameless* (p-value less than 0.01) in Model (3). Blame attribution is also economically important. The odds ratio estimated from Model (3) suggests that firms are *at least four times less likely* to release information following a material negative event for which the firm is likely to be blamed relative to a similar negative event for which the firm is likely to be perceived as blameless.²⁹

In Model (4), we exclude 76 firms that are incorporated outside of the U.S. and continue to find a statistically significant and positive coefficient estimate on *Blameless* (p-value less than 0.01). In terms of economic magnitude, the odds ratio estimated from Model (4) suggests that firms are *at least three times less likely* to release information following a material negative event for which the firm is likely to be blamed relative to a similar negative event for which the firm is likely to be perceived as blameless. We also find that the positive coefficient on *Blameless* in Model (4) is significantly smaller (p-value of 0.02) than the positive coefficient on *Blameless* in

²⁷ The disclosure choice we examine is whether firms provide information about an unexpected negative economic event. This examination is in contrast to a large stream of literature that examines periodic performance reporting choices (e.g., earnings announcements, earnings guidance). We note that many determinants of periodic performance reporting documented in prior research do not explain the disclosure choice after unexpected negative economic events.

²⁸ In untabulated analysis, we also interact *Media coverage* and *Media sentiment* with our blame attribution indicator. The interaction terms are insignificant, suggesting that media coverage and sentiment do not have an incremental association with firm disclosure about blamed negative events.

²⁹ The average disclosure rate is 48 percent for blameless events versus 19 percent for blamed events.

Model (3), suggesting that non-U.S. firms are more sensitive to the reputational and litigation risks than U.S. firms.³⁰

In Model (5), we augment Model (3) using a Tobit duration model (Tobin 1965), in which the duration is the period between the material negative event and the initial disclosure date or 30 days. A positive (negative) coefficient suggests later (earlier) disclosure or longer (shorter) duration. We find that the coefficient estimate on *Blameless* is negative and statistically significant (p-value less than 0.01), suggesting that firms are likely to disclose blameless events earlier after a material negative economic event. In untabulated analysis, we continue to find a negative and statistically significant coefficient estimate on *Blameless* (p-value less than 0.01) when we censor the duration at a lower bound of zero and an upper bound of 30 days.

We further obtain consistent results in the following untabulated robustness tests. First, the coefficient on *Blameless* continues to be positive and statistically significant (p-value less than 0.01) when we estimate Model (3) as a linear probability model. Second, the results continue to hold when we add the absolute value of *Three-day CAR* or use a longer window $([-30, +2])$ relative to the event date to address potential information leakage prior to a lawsuit filing. Third, the statistical significance and economic magnitude remains similar when we exclude 148 events with a non-negative *Three-day CAR* on the event date and when we exclude 95 firms that operate in a regulated industry (e.g., banks, utilities). Taken together, the evidence suggests that perceived blame attribution for a material negative economic event is an important determinant of firms' disclosure decisions beyond event materiality and the firm characteristics.

³⁰ The non-U.S. firms in our sample raised capital from at least one of the three U.S. stock exchanges (NYSE, AMEX, NASDAQ) and are subject to both the Securities Act of 1933 and the Securities Exchange Act of 1934, according to the SEC. Moreover, these non-U.S. firms presumably face similar reputational risks from U.S. stakeholders.

Disclosure within 90 Days

The analysis presented in Table 3 is based on a post-event disclosure window of 30 days following the negative event. In untabulated analyses, we find that 67 percent of the initial event disclosures are released within ten days and 82 percent are released within 15 days after the negative event. This finding suggests that our 30-day post-event period is reasonable for capturing whether or not a firm discloses about the event. However, to incorporate the possibility that firms wait to provide disclosure about the event until the first post-event SEC filing, we also review all 10-K/Q filings in the 90 days after the event. In Table 4, we report the results with this expanded search for disclosures.

Table 4 follows the same columnar structure as Table 3. The pseudo R-squared increases by 7 percent from 0.28 in Model (1) to 0.30 in Model (2) and it further increases by 20 percent from 0.30 in Model (2) to 0.36 in Model (3). Such improvements are attributable to the statistically significant and positive coefficients of *Three-day CAR* (p-value less than 0.01) and Δ *Implied volatility* (p-value of 0.02) in Model (2) and *Blameless* (p-value less than 0.01) in Model (3). For the 90-day window, the odds ratio estimated from Model (3) suggests that firms are *at least five times less likely* to release information following a material negative event for which the firm is likely to be blamed relative to a similar negative event for which the firm is likely to be perceived as blameless.³¹ We continue to document a statistically significant and positive coefficient estimate on *Blameless* (p-value less than 0.01) in Model (4) when we exclude 76 firms that are incorporated outside of the U.S.. The odds ratio estimated from Model (4) suggests that firms are *at least four times less likely* to release information following a material negative event for which the firm is likely to be blamed relative to a similar negative event for which the firm is likely to be

³¹ The average disclosure rate is 67 percent for blameless events versus 28 percent for blamed events.

perceived as blameless.³² Taken together, our inferences remain unchanged after expanding the post-event disclosure window from 30 days to 90 days.

Subsample Tests

We expect our blame attribution hypothesis to apply across a variety of material negative events, and the previous analyses in Tables 3 and 4 provide evidence consistent with this expectation. To assuage concerns that the results are due to the heterogeneity of events included when examining catastrophes, casualty accidents, oil spills, and investor class action lawsuits in the same sample, we conduct tests on subsamples of the material negative events. Table 5 reports separate analyses for the subsample of: a) 192 class action lawsuits in Model (1), b) 191 sudden accidents in Model (2), and c) 87 oil spills that have non-missing data about oil spill specific materiality—oil spill size (in barrels), water depth, and distance to shore—in Model (3).³³ We report only the coefficient and statistical significance for our variable of interest, *Blameless*, and the three spill specific materiality measures because results for the remaining control variables are similar to those reported in Table 3.

Blameless has a positive and statistically significant coefficient for each of the subsamples, with a p-value less than 0.05 in Models (1) and (2), and a p-value less than 0.01 in Model (3). We note that the positive coefficient on *Blameless* in Model (2) for sudden accidents is significantly greater (p-value less than 0.01) than the positive coefficient on *Blameless* in Model (1) for class action lawsuits. This less pronounced effect of blame attribution on disclosure propensity for the lawsuits subsample is consistent with the blame attribution hypothesis given that the lawsuits related to overly optimistic forward-looking statements are endogenously driven by the firm's own

³² The positive coefficient on *Blameless* in Model (4) is significantly smaller (p-value less than 0.01) than the positive coefficient on *Blameless* in Model (3), suggesting that non-U.S. firms are more sensitive to the reputational and litigation risks than U.S. firms.

³³ We exclude industry fixed effects in Model (3) because all firms that experience an oil spill are in the same industry.

past disclosure behavior. However, even for the lawsuit subsample, the odds ratio estimated from Model (1) suggests that firms are *at least three times less likely* to release information following a lawsuit for which the firm is likely to be blamed relative to a lawsuit for which the firm is likely to be perceived as blameless. These subsample analyses suggest that the blame attribution hypothesis applies across different types of material negative economic events.

V. ADDITIONAL ANALYSES

Reputation and Litigation Risks

In Section II, we argue that a firm's disclosure propensity varies across blamed and blameless events because disclosure about blamed events is associated with higher reputation and litigation risks than disclosure about blameless events. That is, stakeholders are more likely to downgrade the firm's reputation or to file a successful lawsuit when the firm issues disclosures about a blamed event than when the firm issues disclosures about a blameless event. We use two reputation proxies and two litigation cost proxies to validate this underlying assertion.

The first reputation proxy is the Fortune 1000 Most Admired Companies score.³⁴ Each year, Fortune selects the largest companies and the largest industries based on sales revenue, and surveys executives, outside directors, and financial analysts regarding the following perceptions of a firm: (a) the ability to attract or retain talents, (b) the quality of management team, (c) the quality of products or services, (d) the degree of innovation, (e) the long-run investment value, (f) the financial health, (g) whether corporate assets are appropriately used, and (h) the extent of social responsibility to the community and environment. Fortune does not disclose a reputation score for all firms, but instead discloses a reputation score when the firm has a reputation score that is among

³⁴ See footnote 5 in Cao, Myers, Myers, and Omer (2015) for a list of studies using the Fortune Most Admired score as a proxy for company reputation.

the top half of its industry and when there are enough firms with available scores in an industry.³⁵ Thus, we are able to identify 126 negative events with at least one publicly available Fortune score during the two years prior to the event and at least one score during the two years after the event. We use these data to calculate the percentage change in the Fortune 1000 Most Admired Companies score in the five years [year -2, year +2] around the event ($\Delta \textit{Fortune}$).

Consistent with our blame attribution hypothesis, only 10 firms disclose after experiencing a blamed event. Such a small sample for the blame-disclose group limits our ability to draw inferences for the interaction term $\textit{Blame} \times \textit{Disclose}$ from a traditional OLS regression because the estimated coefficient is unlikely to approach its expected value. Given that the distribution of $\Delta \textit{Fortune}$ is not skewed and our construction of the 126 Fortune-event sample is not conditional on the blame attribution for the event or whether a firm issues post-event disclosure, we use bootstrapping regressions to infer the population distribution of coefficient estimates (Freedman 1981). The bootstrap method requires independent resampling with replacement because the population of the random sample examined in our previous analysis is analogous to the random sample of the bootstrap resamples (Efron 1979). Specifically, we resample 500 times with replacement because we find limited improvement from using more iterations (Neter, Kutner, Nachtsheim, and Wasserman 1996). We also provide results based on only 200 iterations as a robustness test. We report the mean value of the estimated coefficients and the adjusted R-squared based on resamples and report the statistical significance based on the standard deviation of the resample coefficients.

Our variable of interest is $\textit{Blame} \times \textit{Disclose}$, which captures the impact of the firm disclosing a blamed event on reputation relative to not disclosing a blamed event or disclosing a

³⁵ See the 2014 list as an example at <https://money.cnn.com/magazines/fortune/most-admired/2014/list/>.

blameless event. In Table 6, Models (1)-(3) are based on 500 bootstrapping iterations, and Models (4)-(6) are based on 200 iterations. We first include only the four materiality measures as control variables in Models (1) and (4). However, because $\Delta \textit{Fortune}$ captures the firm's general reputation over a long time period, it is likely to be associated with firm characteristics. We therefore add 25 firm characteristics in Models (2) and (5) and year and industry fixed effects in Models (3) and (6). Results across the six bootstrapping specifications present consistent evidence that firms disclosing a blamed event experience a significant incremental decline in reputation ranging from -2.5 percent to -3.8 percent, compared to firms experiencing a blamed event or disclosing a blameless event (see the coefficients on *Blame* $\times$ *Disclose*). In contrast, firms disclosing a blameless event experience a significant increase in reputation (see the coefficients on *Disclose*). These findings are consistent with the assertion underlying our hypothesis that disclosing a blamed event, but not a blameless event, has a negative impact on the firm's reputation.

One potential concern with relying solely on the Fortune 1000 Most Admired Companies tests is that it represents a long-window comparison, which may capture other events that are unrelated to the material negative economic events in our sample (see, for instance, Andrade, Mitchell, and Stafford (2001) for a similar discussion about long-window comparisons around an M&A transaction). To be sure that our reputation results are not sensitive to this situation, we conduct a shorter-window comparison based on the RavenPack Event Sentiment Score (ESS).³⁶ The advantage of this proxy is that 92 percent (i.e., 354) of our sample events have available data during the pre- and post-event period. As RavenPack's sentiment score is based on the perceptions of business professionals hired by RavenPack to assign a sentiment score to news, the sentiment score captures the firm's reputation. We calculate the average sentiment score across all press

³⁶ Dhaliwal, Goodman, Hoffman, and Schwab (2017) also use this measure to capture firm reputation.

releases and articles with an available ESS score for a firm during the three months prior to and the three months after the negative event (Δ *Sentiment*).

Again, consistent with our blame attribution hypothesis, only 30 firms disclose after a blamed event. Given the small sample size for the blame-disclose group, we use bootstrapping regressions to infer the statistical significance of the interaction term *Blame* $\times$ *Disclose*. Table 7 follows the same columnar structure as Table 6. Across the six bootstrapping specifications, we find consistent evidence that firms disclosing a blamed event experience a significant and incremental decline in reputation ranging from -5.7 percent to -6.8 percent, compared to firms not disclosing a blamed event or disclosing a blameless event (see the coefficients on *Blame* $\times$ *Disclose*). We conclude, given the consistent findings across the Fortune 1000 and RavenPack sentiment scores, that the results support the notion that disclosure of a blamed event is associated with greater reputational costs.³⁷

We next examine the importance of litigation risk in our setting. We use two proxies to capture the remaining litigation risk faced by firms already being sued in our investor class action lawsuit sample. Both litigation cost measures are motivated by Pritchard and Sale (2005). First, we examine the duration of the suit, measured as the number of days between the lawsuit initial filing date and the court final decision date because litigation costs are often based on the number of hours worked by the lawyers. Second, we examine the settlement amount, which is conditional on cases that are not dismissed, with the view that higher amounts suggest greater litigation costs.

Similar to our approach examining reputation cost, we use bootstrapping regressions to infer the statistical significance of the interaction term *Blame* $\times$ *Disclose*, as only 22 firms disclose

³⁷ Without bootstrapping, we find consistent magnitudes and signs of coefficient estimates for the interaction term *Blame* $\times$ *Disclose*. The coefficient on *Blame* $\times$ *Disclose* is -2.8 in the Δ *Fortune* analysis and -6.7 (marginally significant with a two-tailed p-value of 0.15) in the RavenPack analysis.

a blamed lawsuit. Across the six bootstrapping specifications in Table 8, we find consistent evidence that lawsuit duration is significantly longer, ranging from 202 to 293 *more* days, for firms disclosing a blamed lawsuit than for firms not disclosing a blamed lawsuit or firms disclosing a blameless lawsuit. In contrast, firms disclosing a blameless lawsuit spend 80 to 136 *fewer* days to resolve the suit. Blamed lawsuits are, on average, 363 to 450 days longer than blameless lawsuits, consistent with the evidence in Pritchard and Sale (2005).

We next examine the settlement amount for the subset of lawsuits that were not dismissed (i.e., blamed). Table 9 reports results that rely on the same bootstrapping methodology used in the prior analysis of lawsuit duration. We find that firms that disclose pay an additional \$8 million to \$21 million for settlement, which is significantly greater than firms that do not disclose. We conclude that the longer lawsuit duration and the higher settlement amount are consistent with the assertion underlying our hypothesis that disclosure about a blamed case is more likely to help plaintiffs prove their claims compared to a blameless case, leading to higher litigation costs.³⁸

We conduct four sets of tests to assess the robustness of the bootstrapping results presented in Tables 6-9 to controlling for potentially influential observations. First, we use the absolute value of an externally studentized residual greater than two to identify suspected outliers and remove the identified suspected outliers from the sample. Second, we identify observations with a Cook's (1977) distance statistic greater than 0.5 as suspected outliers and remove them from the sample. Third, we estimate robust regressions in which observations with small residuals receive a weight of one and larger residuals receive smaller weights (Huber 1973). The iterative reweighted least squares process continues reweighting the residuals until the sum of less rapidly increasing

³⁸ Without bootstrapping, we find consistent magnitudes and signs of coefficient estimates for the interaction term *Blame* $\times$ *Disclose*. The coefficient on *Blame* $\times$ *Disclose* is 284 (two-tailed p-value less than 0.10) in the lawsuit duration analysis and 10.3 in the settlement amount analysis.

functions of the residuals converges. Finally, we transform the continuous dependent variable to $N/2$ ranks, where N represents the original sample size. In untabulated analyses, we find evidence consistent with the results presented in Tables 6-9, with the exception that the studentized residual robustness test for the lawsuit settlement amount (Table 9) yields an insignificant coefficient for *Disclose* (we find consistent evidence when we use the other three robustness tests). Overall, we conclude that the results presented in Tables 6-9 are not unduly affected by influential observations.

Other Disclosure Characteristics

Our hypothesis speaks to firms' propensity to issue event-specific disclosure, and our empirical tests focus on whether disclosure occurs. In this section we assess the extent to which various other firm disclosure characteristics vary with the blame attribution for the event. These disclosure characteristics are: *timeliness* (measured as how early the firm issues the initial disclosure following the event), *frequency* (measured as how many disclosures the firm provides within the 30 days following the event), *length* (measured as the number of words in the firm's event-specific disclosure), *readability* (measured by the Fog or Bog index of the firm's event-specific disclosure), *tone* (measured as the number of positive words minus the number of negative words divided by the total number of words in the firm's event-specific disclosure, where positive and negative words are based on the Harvard dictionary), *specificity* (measured as the number of words used in specific terms, e.g., locations, organizations, persons, money, percentages, times, dates, divided by the total number of words in the firm's event-specific disclosure, where specific terms are based on the Stanford Named Entity Recognition), *uncertainty* (measured as the number of uncertain words divided by the total number of words in the firm's event-specific disclosure, where uncertain words are based on the Loughran and McDonald (2011) word list), *forward-*

looking information (measured as the number of sentences containing forward-looking terms divided by the total number of sentences in the firm's event-specific disclosure where forward-looking terms are based on the Muslu, Radhakrishnan, Subramanyam, and Lim (2015) and Bozanic, Roulstone and Van Buskirk (2018) word lists), *disclosure channel* (measured as whether the firm disclosure is issued in a press release or an Edgar filing), and *disclosure bundling* (measured as the firm discusses other events in the same disclosure).

Timeliness and frequency are available for the 134 events for which firms disclose, while the remaining disclosure attributes are available for the firms' 209 event-specific disclosures associated with 134 events. Among the 134 events, 102 are blameless and 32 are blamed—note this is consistent with our hypothesis that firms are less likely to issue a disclosure following a blamed event. Among the 209 disclosures, 165 are about blameless events and 44 are about blamed events—note this is again consistent with our hypothesis. The lower propensity for firms to issue event-specific disclosure following a blamed event limits our statistical tests to univariate comparisons of means and medians.

Table 10 reports the results of the univariate comparison of means and medians of the disclosure characteristics. For firms that provide event-specific disclosure, the initial disclosure is released 12 days later for a blamed event than for a blameless event, on average, although this difference is not statistically significant. Firms' blamed event disclosures are marginally significantly (p-value of 0.16) less frequent than blameless event disclosures, on average. Firms' blamed event disclosures are significantly (p-value less than 0.01) shorter (on average 43 words shorter) than those for a blameless event. Although the readability measures do not differ between firms' blamed and blameless event disclosures, firms' blamed event disclosures have a significantly (p-value less than 0.01) more negative tone and contain significantly (p-value of 0.04)

less forward-looking information than firms' blameless event disclosures, on average. We obtain consistent results based on the comparison between the median values with a weaker statistical significance (p-value less than 0.10).

Our prior results suggest that firms are significantly less likely to provide event-specific disclosures following a material negative event for which the firm is likely to be blamed. These univariate results provide suggestive evidence that, when a firm does disclose, the disclosures of blamed events are of shorter length, have a more negative tone, and contain less forward-looking information than firms' event-specific disclosures of blameless events. We view this suggestive evidence as an additional piece of information that is consistent with our blame attribution hypothesis—firms are reluctant to issue event-specific disclosures for material negative economic events for which the firm is likely to be blamed.

Incorporating CEO Career Concerns

Our primary insight is that firms are less likely to disclose information regarding a material negative economic event for which it is likely to be blamed than for a negative economic event for which the firm is likely to be perceived as blameless. We present robust evidence that supports this hypothesis and document the associated impact on firms' reputation and litigation risks. We note, however, that if blamed events convey more negative information to stakeholders than similar blameless events, managers might be more likely to withhold this (bad) news to avoid termination or lower compensation—a situation referred to as the career concern hypothesis (Kothari et al. 2009). We examine whether blame attribution provides incremental information over manager career concerns in explaining firms' disclosure decisions.

To proxy for manager's career concerns, we follow Gibbons and Murphy (1992) and assume that career concerns are weaker when the manager is at a later stage in her career and, thus,

closer to retirement. We use three empirical proxies to capture weaker career concerns: the number of years serving as the firm's CEO (CEO tenure), the number of years the CEO has worked for the firm during her career (CEO-firm tenure), and the CEO's age at the time of the material negative economic event. We use ExecuComp to collect these measures for 212 sample firms and manually collect these measures for the remaining 171 sample firms.

In Table 11, Panels A and B, we present the descriptive statistics of the career concern proxies. As Panel B shows, these variables are significantly correlated and, hence, we augment our main regression model (Model (3) in Table 3 or Table 4) with each career concern proxy at a time. Panel C reports that, consistent with the career concern hypothesis, the coefficient on *Weak career concern* is positive and significant in Models (1) and (3). This suggests managers are more (less) likely to disclose after a material negative economic event when they have weaker (stronger) career concerns. Importantly, *Blameless* remains significantly associated with disclosure propensity across all models, and of similar economic magnitude, indicating that blame attribution is incremental to managers' career concerns in explaining firms' disclosure decisions.³⁹

VI. CONCLUSION AND IMPLICATIONS FOR FUTURE RESEARCH

There is a long-held view of accounting as an information system that reflects underlying economic activities (e.g., Canning 1929; AAA Pathways Commission ⁴⁰). For instance, "...accounting system is a mapping from economic events to financial statements" (De Franco, Kothari, and Verdi 2011, p. 896). We argue that, in the wake of a material negative economic

³⁹ Both the career concern hypothesis and the blame attribution hypothesis assume self-interested managers; however, the blame attribution assumes that the interests of the managers and firm are aligned while the career concern hypothesis assumes that the interests of the managers and firm are misaligned. Specifically, the career concern hypothesis assumes managers want to delay the release of bad news while shareholders demand timely disclosures of bad news. The blame attribution hypothesis assumes that both the manager and the firm are worse off with disclosure of blamed events because stakeholders might use the disclosed facts to downgrade the reputation of the firm and its managers or to bring a lawsuit against the firm and its managers. Therefore, the managers' and the firm's interests are misaligned in the career concern hypothesis but aligned in the blame attribution hypothesis.

⁴⁰ Interpreted from its vision model at <http://commons.aaahq.org/groups/2d690969a3/summary>.

event, a firm's decision to issue event-specific disclosure is affected by whether the firm is likely to be perceived as blamed for the event. We refer to this as the "blame attribution" hypothesis, and present evidence that firms are less likely to provide event-specific disclosure about a material negative event for which the firm is likely to be blamed relative to a negative event for which the firm is likely to be perceived as blameless. We further show that reputation and litigation risks potentially underlie the relation between blame attribution for the event and disclosure propensity, evidenced by a greater decline in firm reputation and higher litigation costs for firms that disclose relative to firms that do not disclose following a blamed event.

In addition to identifying a significant event characteristic—blame attribution—that explains disclosure decisions across a broad spectrum of events (e.g., catastrophes, casualty accidents, oil spills, and investor class action lawsuits), our results have potential implications for studies that use company disclosures (e.g., SEC filings or press releases) to identify economic transactions. In particular, our primary finding that firms are less likely to disclose about a negative event for which the firm is likely to be blamed suggests that, consistent with the discussion in Loughran and McDonald (2016), firm disclosures (i.e., the text of 10-Ks and 8-Ks, the issuance of press releases) capture only a subset of the economic events firms experience.

REFERENCES

- Amir, E., S. Levi, and T. Livne. 2018. Do firms underreport information on cyber-attacks? Evidence from capital markets. *Review of Accounting Studies* 23: 1177–1206.
- Andrade, G., M. Mitchell, and E. Stafford. 2001. New evidence and perspectives on mergers. *Journal of Economic Perspectives* 15 (2): 103–120.
- Baginski, S. P., J. M. Hassell, and M. D. Kimbrough. 2004. Why do managers explain their earnings forecasts? *Journal of Accounting Research* 42 (1): 1–29.
- Baginski, S. P., J. M. Hassell, and W. Hillison. 2000. Voluntary causal disclosures: Tendencies and capital market reaction. *Review of Quantitative Finance and Accounting* 15: 371–389.
- Balakrishnan, K., J. E. Core, and R. S. Verdi. 2014. The relation between reporting quality and financing and investment: Evidence from changes in financing capacity. *Journal of Accounting Research* 52 (1): 1–36.
- Barth, M. E., M. F. McNichols, and G. P. Wilson. 1997. Factors influencing firms' disclosures about environmental liabilities. *Review of Accounting Studies* 2: 35–64.
- Barton, J., and M. Mercer. 2005. To blame or not to blame: Analysts' reactions to external explanations for poor financial performance. *Journal of Accounting and Economics* 39 (3): 509–533.
- Ben-Rephael, A., Z. Da, P. D. Easton, and R. D. Israelsen. 2019. Who pays attention to SEC form 8-K? Working paper, Rutgers University, University of Notre Dame, and Michigan State University.
<https://docs.google.com/viewer?a=v&pid=sites&srcid=ZGVmYXVsdGRvbWFpbmVucmVwaHxneDoxZTQ1NjEzZmVhOWMyODVj>.
- Bird, A., S. A. Karolyi, and T. G. Ruchti. 2017. Political uncertainty and corporate transparency. Working paper, Carnegie Mellon University.
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2986678.
- Bliss, B. A., F. Partnoy, and M. Furchtgott. 2018. Information bundling and securities litigation. *Journal of Accounting and Economics* 65 (1): 61–84.
- Bonsall, S. B., A. J. Leone, B. P. Miller, and K. Rennekamp. 2017. A plain English measure of financial reporting readability. *Journal of Accounting and Economics* 63 (2–3): 329–357.
- Botosan, C. A., and M. S. Harris. 2000. Motivations for a change in disclosure frequency and its consequences: An examination of voluntary quarterly segment disclosures. *Journal of Accounting Research* 38 (2): 329–353.
- Bozanic, Z., D. T. Roulstone, and A. Van Buskirk. 2018. Management earnings forecasts and other forward-looking statements. *Journal of Accounting and Economics* 65 (1): 1–20.
- Brown, S., and S. A. Hillegeist. 2007. How disclosure quality affects the level of information asymmetry. *Review of Accounting Studies* 12: 443–477.
- Brown, T. J., and P. A. Dacin. 1997. The company and the product: Corporate associations and consumer product responses. *Journal of Marketing* 61 (1): 68–84.

- Burks, J. J., C. Cuny, J. Gerakos, and J. Granja. 2018. Competition and voluntary disclosure: Evidence from deregulation in the banking industry. *Review of Accounting Studies* 23: 1471–1511.
- Bushman, R. M., C. D. Williams, and R. Wittenberg-Moerman. 2017. The informational role of the media in private lending. *Journal of Accounting Research* 55 (1): 115–152.
- Canning, J. B. 1929. *The economics of accountancy: A critical analysis of accounting theory*. New York, NY: The Ronald Press Company.
- Cao, Y., J. N. Myers, L. A. Myers, and T. C. Omer. 2015. Company reputation and the cost of equity capital. *Review of Accounting Studies* 20: 42–81.
- Cho, C., R. M. Frankel, and X. Martin. 2019. Law and Lemons. Working Paper, Washington University in St. Louis. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3471434.
- Christensen, H. B., E. Floyd, L. Y. Liu, and M. Maffett. 2017. The real effects of mandated information on social responsibility in financial reports: Evidence from mine-safety records. *Journal of Accounting and Economics* 64 (2–3): 284–304.
- Cook, R. D. 1977. Detection of influential observations in linear regression. *Technometrics* 19 (1): 15–18.
- Coombs, W. T. 2007. Attribution theory as a guide for post-crisis communication research. *Public Relations Review* 33 (2): 135–139.
- De Franco, G., S. P. Kothari, and R. S. Verdi. 2011. The benefits of financial statement comparability. *Journal of Accounting Research* 49 (4): 895–931.
- Dhaliwal, D. S., T. H. Goodman, P. J. Hoffman, and C. M. Schwab. 2017. The incidence, valuation and management of tax-related reputational costs: Evidence from a period of protest. Working Paper, University of Arizona, Purdue University, Indiana University. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2760366.
- Donaldson T., and L. E. Preston. 1995. The Stakeholder theory of the corporation: Concepts, evidence, and implications. *Academy of Management Review* 20 (1): 65–91.
- Dutton, J. E., J. M. Dukerich, and C. V. Harquail. 1994. Organizational images and member identification. *Administrative Science Quarterly* 39 (2): 239–263.
- Dye, R. A. 1985. Disclosure of nonproprietary information. *Journal of Accounting Research* 23 (1): 123–145.
- Efron, B. 1979. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics* 7 (1): 1–26.
- Field, L., M. Lowry, and S. Shu. 2005. Does disclosure deter or trigger litigation? *Journal of Accounting and Economics* 39 (3): 487–507.
- Fombrun, C., and M. Shanley. 1990. What's in a name? Reputation building and corporate strategy. *Academy of Management Journal* 33 (2): 233–258.
- Fombrun, C., and C. V. Riel. 1997. The reputational landscape. *Corporate Reputation Review* 1: 5–13.

- Francis, J., D. Philbrick, and K. Schipper. 1994. Shareholder litigation and corporate disclosures. *Journal of Accounting Research* 32 (2): 137–164.
- Freedman, D. A. 1981. Bootstrapping regression models. *The Annals of Statistics* 9 (6): 1218–1228.
- Fu, R., A. Kraft, and H. Zhang. 2012. Financial reporting frequency, information asymmetry, and the cost of equity. *Journal of Accounting and Economics* 54 (2–3): 132–149.
- Gibbons, R., and K. J. Murphy. 1992. Optimal incentive contracts in the presence of career concerns: Theory and evidence. *Journal of Political Economy* 100 (3): 468–505.
- Griliches, Z., and V. Ringstad. 1970. Error-in-the-variables bias in nonlinear contexts. *Econometrica* 38 (2): 368–370.
- Grundfest, J. A., and P. H. Huang. 2006. The unexpected value of litigation: A real options perspective. *Stanford Law Review* 58: 1267–1336.
- Heitzman, S., C. Wasley, and J. Zimmerman. 2010. The joint effects of materiality thresholds and voluntary disclosure incentives on firms' disclosure decisions. *Journal of Accounting and Economics* 49 (1–2): 109–132.
- Henry, E., and A. J. Leone. 2016. Measuring qualitative information in capital markets research: Comparison of alternative methodologies to measure disclosure tone. *The Accounting Review* 91 (1): 153–178.
- Honigsberg, C., S. Rajgopal, and S. Srinivasan. 2019. The changing landscape of auditor liability. *Journal of Law and Economics* (forthcoming).
- Hope, O., D. Hu, and H. Lu. 2016. The benefits of specific risk-factor disclosures. *Review of Accounting Studies* 21 (4): 1005–1045.
- Huang A., K. W. Hui, and R. Z. Li. 2019. Federal judge ideology: A new measure of ex ante litigation risk. *Journal of Accounting Research* 57 (2): 431–489.
- Huang, Y., R. Jennings, and Y. Yu. 2017. Product market competition and managerial disclosure of earnings forecasts: Evidence from import tariff rate reductions. *The Accounting Review* 92 (3): 185–207.
- Huber, P. J. 1973. Robust regression: Asymptotics, conjectures and Monte Carlo. *Annals of Statistics* 1 (5): 799–821.
- Jung, W., and Y. K. Kwon. 1988. Disclosure when the market is unsure of information endowment of managers. *Journal of Accounting Research* 26 (1): 146–153.
- Khan, M., and R. L. Watts. 2009. Estimation and empirical properties of a firm-year measure of accounting conservatism. *Journal of Accounting and Economics* 48 (2–3): 132–150.
- Kim, I., and D. J. Skinner. 2012. Measuring securities litigation risk. *Journal of Accounting and Economics* 53 (1–2): 290–310.
- Kim, J. -B., Y. Li, and L. Zhang. 2011. Corporate tax avoidance and stock price crash risk: Firm-level analysis. *Journal of Financial Economics* 100 (3): 639–662.
- Kimbrough, M. D., and I. Y. Wang. 2014. Are seemingly self-serving attributions in earnings press releases plausible? Empirical evidence. *The Accounting Review* 89 (2): 635–667.

- Kothari, S. P., S. Shu, and P. D. Wysocki. 2009. Do managers withhold bad news? *Journal of Accounting Research* 47 (1): 241–276.
- Kreps, D. M., and R. Wilson. 1982. Reputation and imperfect information. *Journal of Economic Theory* 27 (2): 253–279.
- Lee, L. F., A. P. Hutton, and S. Shu. 2015. The role of social media in the capital market: Evidence from consumer product recalls. *Journal of Accounting Research* 53 (2): 367–404.
- Leftwich, R. W., R. L. Watts, and J. L. Zimmerman. 1981. Voluntary corporate disclosure: The case of interim reporting. *Journal of Accounting Research* 19: 50–77.
- Li, F. 2008. Annual report readability, current earnings, and earnings persistence. *Journal of Accounting Economics* 45 (2–3): 221–247.
- Loughran, T., and B. McDonald. 2011. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance* 66 (1): 35–65.
- Loughran, T., and B. McDonald. 2016. Textual analysis in accounting and finance: A survey. *Journal of Accounting Research* 54 (4): 1187–1230.
- Madsen, P. M., and Z. J. Rodgers. 2015. Looking good by doing good: The antecedents and consequences of stakeholder attention to corporate disaster relief. *Strategic Management Journal* 36 (5): 776–794.
- Magness, V. 2006. Strategic posture, financial performance and environmental disclosure: An empirical test of legitimacy theory. *Accounting, Auditing & Accountability Journal* 19 (4): 540–563.
- Marinovic, I., and F. Varas. 2016. No news is good news: Voluntary disclosure in the face of litigation. *The RAND Journal of Economics* 47 (4): 822–856.
- McKnight, B., and M. K. Linnenluecke. 2017. Patterns of firm responses to different types of natural disasters. *Business & Society*: 1–28.
- Muslu, V., S. Radhakrishnan, K. R. Subramanyam, and D. Lim. 2015. Forward-looking MD&A disclosures and the information environment. *Management Science* 61 (5): 931–948.
- Neter, J., M. Kutner, W. Nachtsheim, and C. Wasserman. 1996. *Applied linear statistical models*. 4th Edition. McGraw-Hill/Irwin.
- Perry, J., and P. D. Fontnouvelle. 2005. Measuring reputational risk: The market reaction to operational loss announcements. Working paper, Federal Reserve Bank of Boston. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=861364.
- Pritchard A. C., and H. A. Sale. 2005. What counts as fraud? An empirical study of motions to dismiss under the Private Securities Litigation Reform Act. *Journal of Empirical Legal Studies* 2 (1): 125–149.
- Rogers, J. L., and P. C. Stocken. 2005. Credibility of management forecasts. *The Accounting Review* 80(4): 1233–1260.
- Rogers, J. L., D. J. Skinner, and A. Van Buskirk. 2009. Earnings guidance and market uncertainty. *Journal of Accounting and Economics* 48 (1): 90–109.

- Shaver, K. 1985. *The attribution of blame: Causality, responsibility, and blameworthiness*. New York, NY: Springer-Verlag.
- Skinner, D. J. 1994. Why firms voluntarily disclose bad news. *Journal of Accounting Research* 32 (1): 38–60.
- Skinner, D. J. 1997. Earnings disclosures and stockholder lawsuits. *Journal of Accounting Economics* 23 (3): 249–282.
- Tobin, J. 1958. Estimation of relationships for limited dependent variables. *Econometrica* 26 (1): 24–36.
- Verrecchia, R. E. 1983. Discretionary disclosure. *Journal of Accounting and Economics* 5: 179–194.
- Verrecchia, R. E., and J. Weber. 2006. Redacted disclosure. *Journal of Accounting Research* 44 (4): 791–814.
- Walker, K. 2010. A systematic review of the corporate reputation literature: Definition, measurement, and theory. *Corporate Reputation Review* 12 (4): 357–387.
- Weiner, B. 1985. An attributional theory of achievement motivation and emotion. *Psychological Review* 92 (4): 548–573.
- Weiner, B. 1986. *An attributional theory of achievement motivation and emotion*. New York, NY: Springer-Verlag.

APPENDIX A

Importance of Event-specific Disclosures

This appendix provides two examples from our sample of material negative events to shed light on the importance of disclosures about sudden accidents and class action lawsuits to stakeholders.

Sudden accident: General Dynamics, April 2, 2011

On Saturday, April 2, 2011, a General Dynamics (GD) Gulfstream G650 aircraft crashed, killing four test pilots. On Sunday, April 3, the CEO announced the names of the four test pilots, expressing sympathy to their family, friends, and co-workers. The G650 was an important new experimental plane for the company, and it was designed to be Gulfstream's largest, fastest, and most expensive business jet on entry to the market.

Reflecting this importance, on Monday, April 4 (the first trading day after the sudden accident), GD's market-adjusted stock return declined by nearly 2 percent. Moreover, the RavenPack Event Sentiment Score (ESS), which ranges from 0-100 with a score less than (greater than) 50 indicative of financial experts rating the news event as negative (positive), was a 19 for GD's event-specific disclosure. To provide more context, during the 18-year period from 2000 to 2017, GD experienced a similarly low ESS score only when management issued significantly lower than expected earnings guidance or when the firm received significant legal penalties. Thus, the event-specific disclosure was clearly of interest not only to investors but also to a broad group of stakeholders.

Investor class action lawsuit event: Nokia, February 5, 2010

On February 5, 2010, Coughlin, Stoia, Geller, Rudman & Robbins LLP announced a class action lawsuit against Nokia, which issued its own press release on February 8 stating that the suit lacked merit. Nokia's market-adjusted stock return on February 5 declined by nearly 3 percent, with another decline of -1.5 percent on February 8. The RavenPack Event Sentiment Score (ESS), which ranges from 0-100 with a score less than (greater than) 50 indicative of financial experts rating the news event as negative (positive), for both the law firm's announcement and Nokia's event-specific disclosure were clearly negative—22 for the law firm's announcement and 20 for Nokia's event-specific disclosure. To provide more context to the event-specific disclosure, during the 18-year period from 2000 to 2017, Nokia experienced a similarly low ESS score when the firm faced patent infringement legal action or when management announced a substantial loss of product market share. Thus, the event-specific disclosure was clearly of interest to market participants.

APPENDIX B

Examples of Event-specific Disclosures

General Dynamics Chairman and CEO Issues Statement Regarding G650 Test Aircraft Accident

FALLS CHURCH, Va., April 5, 2011 /ENP Newswire/ -- General Dynamics Chairman and Chief Executive Officer Jay L. Johnson issued a statement today regarding the April 2 crash of a Gulfstream G650 test aircraft.

‘Our thoughts and prayers are with the families, friends, and co-workers of our four Gulfstream employees - experimental test pilots Kent Crenshaw and Vivan Ragusa, and technical specialists David McCollum and Reece Ollenburg - who died Saturday in the accident involving the G650 test aircraft in Roswell, New Mexico.’

‘Our sorrow from the loss of these four great men is very deep,’ said Johnson.

He continued, ‘I am confident that as Gulfstream assists aviation authorities in the accident investigation, the cause of this terrible tragedy will be determined. We look forward to continuing the rigorous testing required to achieve flight certification of the aircraft.’

‘The G650 will take its place atop the long line of safe, reliable, high-performance business jets on which Gulfstream has built its superb reputation,’ Johnson said.

General Dynamics, headquartered in Falls Church, Va., employs approximately 90,000 people worldwide. The company is a market leader in business aviation; land and expeditionary combat systems, armaments and munitions; shipbuilding and marine systems; and information systems and technologies.

More information about the company is available online at www.generaldynamics.com.

MELA Sciences Form 8-K Disclosure about an Investor Class Action Lawsuit Event

Item 8.01 — Other Events

On November 19, 2010, a complaint was filed for a purported securities class action lawsuit naming as defendants MELA Sciences, Inc. (the “Company”) and certain of its officers and directors. The lawsuit asserts violations of the Securities Exchange Act of 1934 and alleges among other things that misstatements were made by the Company about its product, MelaFind. The complaint is allegedly filed on behalf of stockholders who purchased the Company’s common stock during the period from February 13, 2009 through November 16, 2010.

The Company believes that it has meritorious defenses and intends to vigorously defend against the lawsuit.

It is possible that additional lawsuits may be filed in the same or other courts making similar allegations as to the same and/or additional defendants. To the extent such similar actions are filed, the Company does not intend to file further Current Reports on Form 8-K describing the additional lawsuits.

APPENDIX C

Variable Definitions

Variable	Definition	Source
Proxies for Blame Attribution		
Blameless	An indicator variable equals 1 when Dismiss or External equals 1, and 0 otherwise	Stanford SCAC
Dismiss	An indicator variable equals 1 for dismissed cases, and 0 for non-dismissed cases.	
External	An indicator variable equals 1 if a sudden accident is caused by external forces, and 0 if a sudden accident is caused by internal forces	
Proxies for Disclosure Propensity		
Disclose (30 days)	An indicator variable equals 1 for firms providing any disclosures regarding a specific material negative event within the month following the event	Edgar, RavenPack, Factiva
Disclose (90 days)	An indicator variable equals 1 for firms providing any disclosures regarding a specific material negative event within the quarter following the event	Edgar, RavenPack, Factiva
Proxies for Event Materiality		
Three-day CAR	Cumulative market-adjusted return during [day 0, day +2] around the initial occurring date of a material negative event	CRSP
Δ Implied volatility	Change in implied volatility derived from option prices [day -3, day +3] around the initial occurring date of a material negative event	Option Metrics
Media coverage	Natural logarithm of the number of news articles about a material negative event on the initial occurring date. Restricted to RavenPack press releases or full articles	RavenPack
Media sentiment	Average sentiment score of news articles about a material negative event on initial occurring date. Sentiment score is transformed as (ESS-50)/50	RavenPack
Firm Characteristics		
Size	Natural logarithm of market capitalization	CRSP
Market-to-book	Natural logarithm of the market-to-book ratio	CRSP, Compustat
Return-on-assets	Income before extraordinary items divided by beginning total assets	Compustat
Loss firm	An indicator variable set to 1 if actual earnings is negative	Compustat
Leverage	Book value of long-term debt divided by beginning total assets	Compustat
Investment cycle	the length of investment cycle is captured by the ratio of depreciation expense to lagged total assets	Compustat

APPENDIX C (CONTINUED)

Variable Definitions

Variable	Definition	Source
Return momentum	Twelve months market adjusted buy-and-hold return in a year	CRSP
Return volatility	Standard deviation of monthly stock returns over a year	CRSP
Firm age	Natural logarithm of the number of years a firm is publicly traded	CRSP
NYSE firm	An indicator variable set to 1 if a firm is listed on NYSE	CRSP
USA firm	An indicator variable set to 1 if a firm is incorporated in the U.S.	Compustat
Litigation industry	An indicator variable set to 1 if a firm operates in a high-litigation industry: biotechnology (SIC codes 2833-2836), computers (3570-3577 and 7370-7374), electronics (3600-3674), and retailing (5200-5961)	Compustat
Litigation RS	An estimated probability of being sued by investors based on Table 8 of Rogers and Stocken (2005)	CRSP
Litigation KS	An estimated probability of being sued by investors based on Table 8 Model (5) of Kim and Skinner (2012)	CRSP, Compustat, Thomson Reuters
Liberal court	An estimated judge ideology based on a firm's primary operating state and the portion of Democratic presidents' appointees in the corresponding circuit	Huang, Hui, and Li [2019]
Herfindahl index	Based on revenues of firms in a SIC three-digit industry	Compustat
Redaction	Natural logarithm of the number of times a firm mentions '24b', '24b-2', 'confidential treatment', 'ct order', 'redact', or 'rule 406' in Exhibit.10 material contracts during the 24 months prior to a material negative event	Edgar
Tariff rate	Annual amount of custom charges divided by the dollar value of imported goods in a SIC three-digit industry	Census data from Prof. Peter Schott's website
Guidance	Natural logarithm of the number of management guidance issued in a year	First Call, IBES
Insider ownership	Percentage of outstanding shares held by insiders, defined as those required to file the Form 4	Thomson Reuters, CRSP
Institutional investor	Percentage of outstanding shares held by institutional investors, defined as those required to file the 13F	Thomson Reuters, CRSP
Analyst following	Natural logarithm of the number of analysts in a year	IBES
Public debt issuance	Natural logarithm of the number of public debt issuances in a year	Thomson Reuters SDC Platinum
Bid-ask Spread	Average of monthly bid-ask spread in a year	CRSP
PIN	Annual probability of informed trade calculated by Brown and Hillegeist [2007]	Prof. Stephen Brown's website
Analyst dispersion	Standard deviation of monthly analysis forecast consensus over a year	IBES

APPENDIX C (CONTINUED)

Variable Definitions

Variable	Definition	Source
Oil Spill Characteristics		
Spill size	Natural logarithm of the number of barrels spilled	BSEE
Water depth	Natural logarithm of the water depth in feet	BSEE
Distance to shore	Natural logarithm of the distance to shore in miles	BSEE
Proxies for Reputation and Litigation Risks		
Δ Fortune	Percentage change in the Fortune 1000 Most Admired Companies score during five years [year -2, year +2] around a material negative event	Forbes
Δ Sentiment	Percentage change in the RavenPack sentiment score based on all press releases and full articles during six months [month -3, month +3] around a material negative event	RavenPack
Lawsuit duration	Number of days between the lawsuit initial filing date and the court final decision date	Stanford SCAC
Lawsuit settlement	Settlement amount in million	Stanford SCAC
Disclosure Characteristics		
Timeliness	Number of days between the material negative event initial occurring date and the first disclosure date	Edgar, Factiva
Frequency	Number of event-specific disclosures issued during the month following a material negative event	Edgar, Factiva
Disclosure CAR	Cumulative market-adjusted return during [day 0, day +2] around a disclosure about the material negative event, based on disclosures issued within the 30 days following the event	CRSP
Length	Number of words in a negative event-specific disclosure	Edgar, Factiva
Fog index	Gunning Fog Index based on texts in a negative event-specific disclosure	Li [2008]
Bog index	Bog Index based on texts in a negative event-specific disclosure	Bonsall et al. [2017]
Tone	(Number of positive words - number of negative words) / total words in a negative event-specific disclosure	Harvard dictionary, Henry and Leone [2016]
Specificity	Number of words in specific terms (locations, organizations, persons, money, percentages, times, dates) divided by total number of words in a negative event-specific disclosure	Stanford NER, Hope et al. [2016]
Uncertainty	Number of uncertain words divided by total number of words in a negative event-specific disclosure	Loughran and McDonald [2011]
Forward-looking	Number of sentences containing forward-looking terms divided by total number of sentences in a negative event-specific disclosure	Muslu et al. [2015], Bozanic et al. [2018]
PR vs. Edgar	An indicator variable set to 1 if a negative event-specific disclosure is issued in a press release, and 0 if issued in an Edgar filing	Edgar, Factiva
Bundled	An indicator variable set to 1 if events other than the material negative event in our sample are discussed in an event-specific disclosure, and 0 otherwise	Edgar, Factiva

TABLE 1
Summary Statistics of Variable Distributions

This table presents descriptive statistics of the variables used in the analysis. Please refer to Appendix C for variable definitions.

	N	Mean	25th	Median	75th	Std Dev
Blameless	383	0.55	0.00	1.00	1.00	0.50
Dismiss	192	0.56	0.00	1.00	1.00	0.50
External	191	0.54	0.00	1.00	1.00	0.50
Disclose (30 days)	383	0.35	0.00	0.00	1.00	0.48
Disclose (90 days)	383	0.50	0.00	0.00	1.00	0.50
Three-day CAR	383	-0.02	-0.04	-0.01	0.01	0.08
Δ Implied volatility	383	0.01	-0.01	0.00	0.02	0.06
Media coverage	383	1.67	1.39	1.67	1.95	0.51
Media sentiment	383	-0.40	-0.54	-0.40	-0.32	0.15
Size	383	7.70	6.29	7.98	9.70	2.55
Market-to-book	383	-0.45	-1.04	-0.47	0.00	1.20
Return-on-assets	383	0.03	0.00	0.04	0.10	0.20
Loss firm	383	0.21	0.00	0.00	0.00	0.41
Leverage	383	0.16	0.00	0.12	0.26	0.17
Investment cycle	383	0.05	0.02	0.04	0.07	0.07
Return momentum	383	0.27	-0.05	0.15	0.36	0.96
Return volatility	383	0.12	0.06	0.09	0.15	0.12
Firm age	383	2.51	1.79	2.71	3.61	1.30
NYSE firm	383	0.62	0.00	1.00	1.00	0.49
USA firm	383	0.80	1.00	1.00	1.00	0.40
Litigation industry	383	0.22	0.00	0.00	0.00	0.41
Litigation RS	383	-2.43	-2.64	-2.41	-2.23	0.31
Litigation KS	383	0.40	0.17	0.39	0.54	0.26
Liberal court	383	0.35	0.18	0.30	0.58	0.20
Herfindahl index	383	0.12	0.05	0.08	0.13	0.13
Redaction	383	0.36	0.00	0.00	0.00	1.11
Tariff rate	383	0.001	0.000	0.000	0.001	0.006

TABLE 1 (CONTINUED)
Summary Statistics of Variable Distributions

This table presents descriptive statistics of the variables used in the analysis. Please refer to Appendix C for variable definitions.

	N	Mean	25th	Median	75th	Std Dev
Guidance	383	0.77	0.00	0.00	1.61	0.86
Insider ownership	383	0.03	0.00	0.00	0.00	0.32
Institutional investor	383	0.53	0.22	0.63	0.83	0.35
Analyst following	383	2.08	0.69	2.56	3.22	1.35
Public debt issuance	383	0.11	0.00	0.00	0.00	0.40
Bid-sak Spread	383	0.37	0.01	0.03	0.06	4.64
Analyst dispersion	383	1.30	0.02	0.16	0.40	18.59
Spill size	87	5.48	4.62	5.20	6.35	1.13
Water depth	87	6.94	5.46	7.25	8.43	1.71
Distance to shore	87	4.17	3.83	4.37	4.79	0.87
Δ Fortune	126	-1.36	-8.91	-0.92	5.23	10.08
Δ Sentiment	354	-3.32	-13.74	-3.25	5.99	18.39
Lawsuit duration	192	993	574	903	1286	600
Lawsuit settlement	84	24.0	2.9	6.0	23.8	56.2
Timeliness	134	11.2	2.0	5.0	14.0	33.4
Frequency	134	1.56	1.00	1.00	2.00	1.10
Disclosure CAR	209	-0.01	-0.03	-0.01	0.02	0.07
Length	209	162	95	140	196	113
Fog index	209	18.09	14.22	16.28	19.27	8.74
Bog index	209	62.81	51.00	63.00	75.00	17.34
Tone	209	-0.06	-0.10	-0.05	-0.02	0.05
Specificity	209	0.37	0.29	0.36	0.44	0.10
Uncertainty	209	0.02	0.01	0.01	0.03	0.02
Forward-looking	209	0.25	0.11	0.25	0.36	0.18
PR vs. Edgar	209	0.49	0.00	0.00	1.00	0.50
Bundled	209	0.26	0.00	0.00	1.00	0.44

TABLE 2
Correlation Matrix of Selected Variables

Please refer to Appendix C for detailed variable definitions. Correlations in bold refer to a two-tailed significance level less than 0.10.

Right: Pearson Left: Spearman	Blameless	Disclose (30 days)	Disclose (90 days)	Three-day CAR	Δ Implied volatility	Media coverage	Media sentiment
Blameless		0.31	0.39	0.06	0.08	0.04	0.05
Disclose (30 days)	0.31		0.74	0.12	0.02	-0.11	-0.06
Disclose (90 days)	0.39	0.74		0.04	0.12	-0.08	0.05
Three-day CAR	-0.04	0.04	-0.04		-0.34	0.01	0.06
Δ Implied volatility	0.11	0.03	0.10	-0.23		-0.03	-0.04
Media coverage	0.06	-0.11	-0.08	0.02	-0.03		0.27
Media sentiment	0.04	-0.10	-0.01	0.11	-0.01	0.32	

TABLE 3

Disclosure Propensity within the *Month* following a Material Negative Economic Event

This table presents results from logistic regressions. The dependent variable is *Disclose (30 days)*, an indicator variable for firms providing any disclosure regarding a specific negative economic event within the month following the event. Models (1)-(3) examine the full sample, while Model (4) excludes non-U.S. firms. Model (5) examines a Tobit duration model (Tobin 1965), in which the duration is the period between the material negative event and the initial disclosure date or 30 days. All firm-level determinants are from the previous year. Please refer to Appendix C for variable definitions. * Two-tailed $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

	Model (1)	Model (2)	Model (3)	Model (4) USA only	Model (5) Duration
Blameless			1.40 ***	1.19 ***	-15.70 ***
Three-day CAR		5.95 ***	6.22 ***	6.40 ***	-66.51 ***
Δ Implied volatility		3.51	1.86	2.64	-6.95
Media coverage		-0.23	-0.49	-0.61	4.45
Media sentiment		-1.51	-1.34	-1.15	19.30 *
Size	-0.09	-0.06	-0.01	0.11	-0.08
Market-to-book	-0.09	-0.02	0.14	0.51	-0.46
Return-on-assets	1.29	1.93 *	1.67	1.88	-13.39
Loss firm	0.87 *	1.09 *	1.09 *	1.19 *	-8.49
Leverage	-0.26	-0.22	0.27	0.79	-1.94
Investment cycle	1.92	1.70	0.42	0.68	-7.77
Return momentum	-0.09	-0.26	-0.35	-0.21	4.81
Return volatility	-2.25	-2.45	-3.16	-3.44	32.60
Firm age	0.13	0.16	0.13	0.08	-1.48
NYSE firm	-0.12	-0.31	-0.41	-0.58	1.87
USA firm	-0.03	0.05	0.07		0.26
Litigation industry	0.17	0.16	0.18	0.12	-5.50
Litigation RS	-0.09	0.05	0.30	0.45	2.13
Litigation KS	0.22	0.33	0.23	0.48	0.25
Liberal court	-0.81	-0.71	-0.70	-0.59	7.13
Herfindahl index	0.36	0.45	1.18	1.02	-3.70
Redaction	-0.07	-0.04	-0.01	0.02	-0.57
Tariff rate	3.81	1.88	-0.71	-0.58	100.15
Guidance	-0.24	-0.21	-0.17	-0.26	1.72
Insider ownership	1.65	2.97	4.89 **	5.91 ***	-57.26 **
Institutional investor	-0.59	-0.52	-0.32	0.02	2.96
Analyst following	0.23	0.21	0.09	0.10	-0.83
Public debt issuance	-0.30	-0.35	-0.25	-0.53	2.65
Bid-ask Spread	-1.85 *	-1.71	-1.60	-1.50	18.22
Analyst dispersion	-0.05	-0.04	0.11	-0.27	-1.37
Fixed effects	year, industry	year, industry	year, industry	year, industry	year, industry
Std. error clustered on	firm	firm	firm	firm	
N	383	383	383	307	383
Pseudo R-squared	0.17	0.20	0.25	0.22	NA

TABLE 4

Disclosure Propensity within the *Quarter* following a Material Negative Economic Event

This table presents results from logistic regressions. The dependent variable is *Disclose (90 days)*, an indicator variable for firms providing any disclosure regarding a specific negative economic event within the quarter following the event. Models (1)-(3) examine the full sample, while Model (4) excludes non-U.S. firms. All firm-level determinants are from the previous year. Please refer to Appendix C for variable definitions. * Two-tailed $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

	Model (1)	Model (2)	Model (3)	Model (4) USA only
Blameless			1.62 ***	1.42 ***
Three-day CAR		5.70 ***	5.15 **	4.15 **
Δ Implied volatility		5.51 **	4.26 *	3.39
Media coverage		0.36	0.11	0.10
Media sentiment		-0.26	0.09	0.74
Size	-0.08	-0.12	-0.05	0.11
Market-to-book	-0.31 *	-0.34 *	-0.19	0.14
Return-on-assets	-0.29	0.16	-0.48	-0.32
Loss firm	0.56	0.76	0.70	0.82
Leverage	-0.65	-0.28	0.17	0.39
Investment cycle	0.33	-0.86	-2.05	-1.02
Return momentum	-0.30	-0.41	-0.51	-0.44
Return volatility	-2.84	-2.76	-3.78	-3.59
Firm age	0.09	0.07	0.01	-0.07
NYSE firm	-0.31	-0.37	-0.56	-0.70
USA firm	0.72	0.83 *	0.86 *	
Litigation industry	-0.33	-0.27	-0.29	-0.53
Litigation RS	0.54	0.54	0.97	1.15
Litigation KS	1.15	1.38	1.17	1.32
Liberal court	-0.45	-0.25	-0.29	-0.95
Herfindahl index	0.13	0.13	0.94	0.70
Redaction	-0.07	-0.03	0.02	0.04
Tariff rate	-2.89	-3.17	-7.47	-14.15
Guidance	-0.34	-0.38 *	-0.32	-0.45 *
Insider ownership	-0.63	-0.25	1.85	2.88
Institutional investor	-0.30	-0.08	0.17	0.96
Analyst following	0.10	0.09	-0.04	-0.26
Public debt issuance	-0.49	-0.43	-0.32	-0.50
Bid-ask Spread	-1.32	-1.13	-0.99	-0.98
Analyst dispersion	-0.10	-0.09	0.08	-0.31
Fixed effects	year, industry	year, industry	year, industry	year, industry
Std. error clustered on	firm	firm	firm	firm
N	383	383	383	307
Pseudo R-squared	0.28	0.30	0.36	0.32

TABLE 5
Subsample Tests

This table presents results from logistic regressions. The dependent variable is *Disclose (30 days)*, an indicator variable for firms providing any disclosure regarding a specific negative economic event within the month following the event. Model (1) is based on a sample that includes 192 investor class action lawsuits claiming overly optimistic forward-looking statements. Model (2) is based on a sample that includes 191 sudden accidents (61 catastrophes, 41 casualty accidents, and 89 oil spills). Model (3) is based on a sample that includes 87 oil spills with available spill size, water depth, and distance to shore. All control variables in Table 3 are included but unreported for brevity. Please refer to Appendix C for variable definitions. * Two-tailed $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

	Model (1)	Model (2)	Model (3)
Sample:	Lawsuits	Sudden Accidents	Oil Spills
Blameless	1.05 **	3.15 **	11.36 ***
Spill size			2.36 ***
Water depth			1.07 **
Distance to shore			-2.45 **
Control variables	Yes	Yes	Yes
Fixed effects	year, industry	year, industry	year
Std. error clustered on	firm	firm	firm
N	192	191	87
Pseudo R-squared	0.24	0.60	0.99

TABLE 6
Change in Fortune 1000 Most Admired Companies Score

This table presents bootstrapping OLS regressions of the percentage change in the Fortune 1000 Most Admired Companies score in the five years [year -2, year +2] around a negative economic event on the interaction term *Blame* × *Disclose*, *Blame*, and *Disclose*. *Blame* is the opposite indicator to *Blameless*. *Disclose* refers to *Disclose (30 days)*. Models (1) and (4) include the four event materiality measures in the regression. Models (2) and (5) further include 25 firm characteristics. Models (3) and (6) additionally include year and industry fixed effects. The original sample includes 126 negative events with at least one publicly available Fortune score during the two years prior to the event and at least one score during the two years after the event. Due to the small sample size, especially only 10 firms disclosing after experiencing a blamed event, we bootstrap 500 or 200 iterations to generate a distribution for estimated coefficients and the adjusted R-squared. Models (1)-(3) are based on 500 iterations and Models (4)-(6) are based on 200 iterations. Control variables are not reported for brevity. Please refer to Appendix C for detailed variable definitions. * Two-tailed p<0.10; ** p<0.05; *** p<0.01.

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)
Bootstrap Iterations:	500	500	500	200	200	200
Blame × Disclose	-2.7 ***	-3.8 ***	-3.0 ***	-2.9 ***	-3.4 ***	-2.5 ***
Blame	-0.4 ***	0.5 ***	2.9 ***	-0.5 **	0.5 ***	2.6 ***
Disclose	2.0 ***	3.5 ***	3.7 ***	2.2 ***	3.3 ***	3.5 ***
Event materiality	Yes	Yes	Yes	Yes	Yes	Yes
Firm characteristics	No	Yes	Yes	No	Yes	Yes
Fixed effects	No	No	Yes	No	No	Yes
Adj. R-squared	0.07	0.38	0.59	0.07	0.37	0.58

TABLE 7
Change in RavenPack Sentiment Score

This table presents bootstrapping OLS regressions of the percentage change in the RavenPack average sentiment score during the six months [month -3, month +3] around a negative economic event on the interaction term *Blame* × *Disclose*, *Blame*, and *Disclose*. *Blame* is the opposite indicator to *Blameless*. *Disclose* refers to *Disclose (30 days)*. Models (1) and (4) include the four event materiality measures in the regression. Models (2) and (5) further include 25 firm characteristics. Models (3) and (6) additionally include year and industry fixed effects. The original sample includes 354 negative events that have at least one press release or full article with available ESS score in RavenPack during the three months prior to the event and at least one press release or full article with available ESS score in RavenPack during the three months after the event. Due to the small sample size, especially our interest group of Blame-Disclose, we bootstrap 500 or 200 iterations to generate a distribution for estimated coefficients and the adjusted R-squared. Models (1)-(3) are based on 500 iterations and Models (4)-(6) are based on 200 iterations. Control variables are not reported for brevity. Please refer to Appendix C for detailed variable definitions. * Two-tailed $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)
Bootstrap Iterations:	500	500	500	200	200	200
Blame × Disclose	-6.8 ***	-6.1 ***	-6.0 ***	-6.7 ***	-5.9 ***	-5.7 ***
Blame	0.2	-1.9 ***	-1.0 ***	0.1	-1.8 ***	-1.0 ***
Disclose	0.3 **	-0.4 ***	0.1	0.1	-0.6 ***	-0.2
Event materiality	Yes	Yes	Yes	Yes	Yes	Yes
Firm characteristics	No	Yes	Yes	No	Yes	Yes
Fixed effects	No	No	Yes	No	No	Yes
Adj. R-squared	0.04	0.14	0.18	0.04	0.13	0.17

TABLE 8
Lawsuit Duration

This table presents bootstrapping OLS regressions of the lawsuit duration, measured as the number of days between the lawsuit initial filing date and the court final decision date, on the interaction term *Blame* × *Disclose*, *Blame*, and *Disclose*. *Blame* is the opposite indicator to *Blameless*. *Disclose* refers to *Disclose* (30 days). Models (1) and (4) include the four event materiality measures in the regression. Models (2) and (5) further include 25 firm characteristics. Models (3) and (6) additionally include year and industry fixed effects. The original sample includes 192 investor class action lawsuits claiming overly optimistic forward-looking statements. Due to the small sample size, especially our interest group of Blame-Disclose, we bootstrap 500 or 200 iterations to generate a distribution for estimated coefficients and the adjusted R-squared. Models (1)-(3) are based on 500 iterations and Models (4)-(6) are based on 200 iterations. Control variables are not reported for brevity. Please refer to Appendix C for detailed variable definitions. * Two-tailed p<0.10; ** p<0.05; *** p<0.01.

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)
Bootstrap Iterations:	500	500	500	200	200	200
Blame × Disclose	293 ***	279 ***	202 ***	284 ***	281 ***	203 ***
Blame	363 ***	444 ***	450 ***	373 ***	436 ***	446 ***
Disclose	-136 ***	-80 ***	-99 ***	-129 ***	-93 ***	-110 ***
Event materiality	Yes	Yes	Yes	Yes	Yes	Yes
Control variables	No	Yes	Yes	No	Yes	Yes
Fixed effects	No	No	Yes	No	No	Yes
Adj. R-squared	0.24	0.40	0.46	0.24	0.40	0.45

TABLE 9
Settlement Amount for Blamed Lawsuits

This table presents bootstrapping OLS regressions of the lawsuit settlement amount (in million) on *Disclose* because the sample includes only blamed events. *Disclose* refers to *Disclose (30 days)*. Models (1) and (4) include the four event materiality measures in the regression. Models (2) and (5) further include 25 firm characteristics. Models (3) and (6) additionally include year and industry fixed effects. The original sample includes 84 (blamed) settled investor class action lawsuits claiming overly optimistic forward-looking statements. Due to the small sample size, especially our interest group of Blame-Disclose, we bootstrap 500 or 200 iterations to generate a distribution for estimated coefficients and the adjusted R-squared. Models (1)-(3) are based on 500 iterations and Models (4)-(6) are based on 200 iterations. Control variables are not reported for brevity. Please refer to Appendix C for detailed variable definitions. * Two-tailed $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)
Bootstrap Iterations:	500	500	500	200	200	200
Disclose	8.0 ***	10.0 ***	19.6 ***	11.1 ***	9.8 ***	20.9 ***
Event materiality	Yes	Yes	Yes	Yes	Yes	Yes
Control variables	No	Yes	Yes	No	Yes	Yes
Fixed effects	No	No	Yes	No	No	Yes
Adj. R-squared	0.18	0.46	0.66	0.19	0.45	0.65

TABLE 10
Disclosure Characteristics

This table presents the mean and median values of timeliness and frequency for the 134 events for which firms disclose and the mean and median values of the remaining disclosure characteristics for the 209 event-specific disclosures associated with the 134 events. We classify events or disclosures into two groups based on the blame attribution indicator *Blameless*. Please refer to Appendix C for detailed variable definitions.
* Two-tailed $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

	Mean			Median		
	Blameless	Blamed	Diff in Mean	Blameless	Blamed	Diff in Median
Timeliness	8.35	20.25	11.90	5.50	4.50	-1.00
Frequency	1.62	1.38	-0.24	1.00	1.00	0.00
Length	171.53	128.18	-43.35 ***	144.00	123.50	-20.50 *
Fog index	18.23	17.59	-0.64	16.48	15.72	-0.76
Bog index	62.63	63.43	0.80	62.00	63.50	1.50
Tone	-0.05	-0.09	-0.03 ***	-0.05	-0.10	-0.05 *
Specificity	0.36	0.37	0.01	0.36	0.39	0.03
Uncertainty	0.02	0.02	0.00	0.01	0.01	0.00
Forward-looking	0.27	0.20	-0.06 **	0.25	0.19	-0.06 *
PR vs. Edgar	0.51	0.43	-0.08	1.00	0.00	-1.00
Bundled	0.25	0.32	0.07	0.00	0.00	0.00

TABLE 11
CEO Career Concern

This table presents the analysis on CEO career concern. We use three empirical proxies to capture CEO's weak career concern: the number of years serving as the firm's CEO (CEO tenure), the number of years the CEO has worked for the firm during her career (CEO-firm tenure), and the CEO's age at the time of the material negative economic event. Panel A presents descriptive statistics of the weak career concern variables. Panel B presents the correlations among the weak career concern variables and all correlations are significant with a p-value less than 0.01. Panel C presents results from logistic regressions. We augment the Model (3) in Tables 3-4 by incorporating one empirical proxy for CEO weak career concern at a time. Specifically, Models (1) and (2) use CEO tenure as a proxy for weak career concern, Models (3) and (4) use CEO-firm tenure as a proxy for weak career concern, and Models (5) and (6) use CEO age as a proxy for weak career concern. Models (1), (3), and (5) examine disclosure propensity within the month following a material negative economic event, i.e., the dependent variable is *Disclose (30 days)*. Models (2), (4), and (6) examine disclosure propensity within the quarter following a material negative economic event, i.e., the dependent variable is *Disclose (90 days)*. The same set of control variables and model specifications are included as those in Model (3) of Tables 3-4 (i.e., 25 firm characteristics, four event materiality measures, industry and year fixed effects, and clusters standard errors by firm). We report only the coefficients of *Blameless* and *Weak Career Concern* for brevity. *Weak Career Concern* takes the log transformation of CEO tenure, CEO-firm tenure, or CEO age. * Two-tailed $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

PANEL A						
	N	Mean	25th	Median	75th	Std Dev
CEO Tenure	381	7	2	5	10	8
CEO-Firm Tenure	270	17	6	13	26	13
CEO Age	374	55	51	56	60	8

PANEL B			
Right: Pearson Left: Spearman	CEO Tenure	CEO-Firm Tenure	CEO Age
CEO Tenure		0.46	0.34
CEO-Firm Tenure	0.50		0.37
CEO Age	0.28	0.35	

TABLE 11 (CONTINUED)
CEO Career Concern

This table presents the analysis on CEO career concern. We use three empirical proxies to capture CEO's weak career concern: the number of years serving as the firm's CEO (CEO tenure), the number of years the CEO has worked for the firm during her career (CEO-firm tenure), and the CEO's age at the time of the material negative economic event. Panel A presents descriptive statistics of the weak career concern variables. Panel B presents the correlations among the weak career concern variables and all correlations are significant with a p-value less than 0.01. Panel C presents results from logistic regressions. We augment the Model (3) in Tables 3-4 by incorporating one empirical proxy for CEO weak career concern at a time. Specifically, Models (1) and (2) use CEO tenure as a proxy for weak career concern, Models (3) and (4) use CEO-firm tenure as a proxy for weak career concern, and Models (5) and (6) use CEO age as a proxy for weak career concern. Models (1), (3), and (5) examine disclosure propensity within the month following a material negative economic event, i.e., the dependent variable is *Disclose (30 days)*. Models (2), (4), and (6) examine disclosure propensity within the quarter following a material negative economic event, i.e., the dependent variable is *Disclose (90 days)*. The same set of control variables and model specifications are included as those in Model (3) of Tables 3-4 (i.e., 25 firm characteristics, four event materiality measures, industry and year fixed effects, and clusters standard errors by firm). We report only the coefficients of *Blameless* and *Weak Career Concern* for brevity. *Weak Career Concern* takes the log transformation of CEO tenure, CEO-firm tenure, or CEO age. * Two-tailed p<0.10; ** p<0.05; *** p<0.01.

PANEL C						
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)
Proxy for Weak Career Concern:	CEO Tenure	CEO Tenure	CEO-Firm Tenure	CEO-Firm Tenure	CEO Age	CEO Age
Post-Event Disclosure Window:	30 Days	90 Days	30 Days	90 Days	30 Days	90 Days
Blameless	1.42 ***	1.64 ***	0.98 **	1.10 **	1.41 ***	1.71 ***
Weak career concern	0.27 **	0.05	0.59 **	0.42	0.84	0.51
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	year, industry	year, industry	year, industry	year, industry	year, industry	year, industry
Std. error clustered on	firm	firm	firm	firm	firm	firm
N	381	381	270	270	374	374
Pseudo R-squared	0.26	0.37	0.28	0.41	0.27	0.39